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Wu et al.

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(45) **Date of Patent:** **Sep. 9, 2025**

(54) **MOLECULAR MARKER COMBINATION
LINKED TO QUANTITATIVE TRAITS OF
TEA PLANT CAFFEINE CONTENT**

(58) **Field of Classification Search**
None
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1104 days.

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§ 371 (c)(1),

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(51) **Int. Cl.**
C12Q 1/68 (2018.01)

C12Q 1/6827 (2018.01)

(52) **U.S. Cl.**
CPC **C12Q 1/6827** (2013.01); **C12Q 2600/156**
(2013.01)

(57) **ABSTRACT**

A molecular marker combination linked to quantitative traits of tea plant caffeine content, including a SNP site 1, a SNP site 2, a SNP site 3, a SNP site 4, a SNP site 5 and a SNP site 6, which are located in tea genomes Scaffold4239:309117, Scaffold115:803980, Scaffold720:596655, Scaffold3614:66549, Scaffold349:3413816 and Scaffold920:281727, respectively, and genotypes thereof are extremely significantly correlated with the caffeine content is provided. A detection method for detecting each site, and one or more molecular marker site is used to evaluate the tea plant caffeine content.

1 Claim, 18 Drawing Sheets

Specification includes a Sequence Listing.

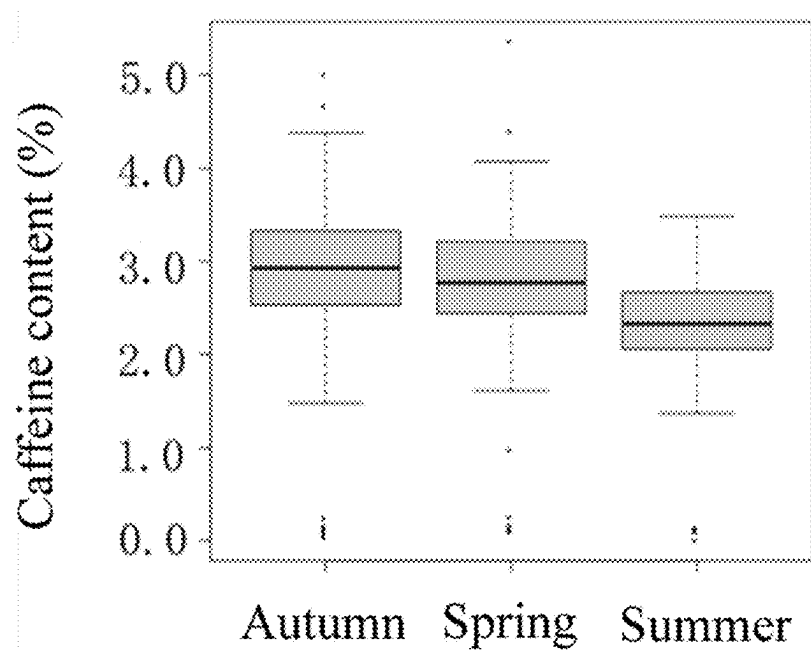


FIG. 1

Scaffold4239:309117 (as shown in SEQ ID NO:1)

GAAGGCTCTGGAGTAGCTGAAGTTGTTATGAGCTTGTCTAGGCCGAAATCA
GCCAGGTGAGCTTCAAAATCGGCGTCAATAGGACGTTCTGAGGCTTGAC
ATCGCCATGAACCATGGCGGTGGAGTGGAGGAAGGCGAGGCCGCGGGCGA
TTCCGAGGGCTATTAGGTGGCGCATTGGCCAATTCAATACATGCCCCGTCTG
GTGAGAAGCTTCTTGAAGCAATGTGGCTAGGTTTCCGTTAGGCATATAGTC
GTAGACTAAGAGTCTGAGGTCTGGTGGTCCGGCGAAGTACCCACGGAGGA
CTGTGAGGTTTCTGTGCTTCACTCTCCCGAGCGATTCCGCTTCTTTTCTGAA
CATGTTTTCGTCTAGCGATCCATCAGGGAGTCTCCGAATCGAAAGCACCAATT
CCATCACTGTAAACAGGCTTTGAAGACTAACCCTATCGAGTCCTGCTTAG
AACGTTCTCTTCATCGAATTGTCTCGTTGCTTCGGTTGTTTCNGCTAGAGTG
ATCTTGTTATTGAACATAACAAGCTTTGGACCGCCATTATCGCCACTTCCAC
GACCTCCGCTGGCTGCAGCTGAGCTTGCTCTTGCTGGGCTGCGCTTTTTCT
CTCCGGCAGCCTTTTTCTTGAGCCTCTTGCGCCACCGCAAGAGACTGTAA
GTTGTAGAAGCAACAACACAGTGCTAAGAGGAAACCACCACTAACAGCCA
TGGCAATAAACATGATCAGCCTCTTCTTCTATTACTCATCTCTTCGCAATTTC
GTGCTTAAGGGTTTCCACATAAGTTCGGATTTCCTGCATAATCAGATGGAT
CGTTGAATCTTGAAGCCAGCATTTGTTGGAATCTCGCCGAGAGGTTGTTTT
GGGATACATTGAAGTAGACCAAGCTAGAGATGAGTGAAATGTTTGCTGGAA
TCGGTCCGGTCAGGTTGTTTGCAGAGAGATTGAGGACTGTGAGGTTTGATA
AATTGGACAATGAGTCTGGTATTTGGCCTGG

upstream primer

(as shown in SEQ ID NO:2)

downstream primer

(as shown in SEQ ID NO:19)

FIG. 2

Scaffold115:803980 (as shown in SEQ ID NO:4)

AATCATTAAAGAGTCATTATGGTAATCATGAGCTTAATTACTCCAAGTAAAGC
CAATCTTCATCATAGAAATAAAATTACAAAAAAAAAAAAAAAAAAAAAG
TCTTTCAGCTGAACAACCCATCCCTGCAACTGCACCAACCATAATTGAGATCT
AAATCTGAAGGAACTTGCTTGAGATCTAAATCTGAAGGAACTTGCTTGCTT
AGGAACATCCACATCCATGATTTCTACAATTTTGGGAAGACACAGAACCAG
AGAAGATGACTCAAAATCAAGCAGCAATTGTAAAGAAAATTCGACCAATCG
AAATCATCTTGGAATTAATCATTTGTAGCCTCCTTCATCTCCACCACACTTC
TCCTCCTACTTCCATGCGATTACGTCGACGGCAGCCCTATTCCCAACATCATA
TTCAAAGGACTCCCTCCACCTTCCACGCCCTTCGTCGTCCTCCCTCATCTTCG
CCTTCTCCGGAGCCTTGAGCGCCTTGTTGATCCACGACATCCCTCTTTGC
CAAGCTCTGCGAGTTCTCTTCCATGGCCTCCATGACCTCTGCTCTCTCTT
GCTACTTTGGGCTATGTTCTTCACCTGTTTCAACCACAACCCAGGTAAAA
CTCGAATTCAGACATCACATGGTAAGAAAACAAGTTATTAAGGTTTTTAACC
TTATAAGACTTTTTTCTTTTTCTTTTCTTCTCCTTCTGTCACGACACGTGG
TGTTGTTTTAAATTAATAAATCGTGATCAGATATGGATATACAATCGCGTGG
TCAGTTGAAATTACTATTGGTATGCTTTATAATACCGTGTGTTGTGTAATA
AACTTGTTTTGTGATGTTGTTGGTCTGTTATGTACTTGGTGTGTTGAAAT
AATATTACATAAATTGAATAAGCCTTTATTATGTGGAGATCCGATGGATTA
ATGATGCATATTGTACAGAATTCAAATGATTTCAATTTGAGCATGGTGAC
GAGGGTTCCAAGCCCTG

upstream primer
(as shown in SEQ ID NO:5)

downstream primer
(as shown in SEQ ID NO:20)

FIG. 3

Scaffold720:596655 (as shown in SEQ ID NO:7)

GCCCACTAGATAATGGCAAAACCATCTTTTCTAAAGCCCACTTGAGGGTGG
CAAAATCTGCATTTTAGCCTACTTGCGAGTAGCAAACCATTTTCAGAACAA
ACATGAAATGTTTTGTTCCAATTIAGGCGATTGGTCTACACTTGTTTGCTG
TGCGCATAACATAGTAATTC TAGGCTAATAACCGGAATAAAAGCATATAATG
TTTTCAAAACAAGATATCATGGTTAGTAGAGGCCGATCTTTGATCGGGCGA
GTGGGTATGCTCATAGCTGACCTGCCCAATTGTACCCAAGTTCCCGAGCC
GAGACAACTTGGTGATGACGGACACGTGCCCTTTGGCACCGAGTCAGG
GTTTCTTAAATCTCCCAAAATAAAATATTTAGGTGGCGACTCTGTATCTGG
CAAAGCAGTCCATGTTTGGCACTTCTTTTCTAAAAATCTGTTTTCTCAAAAC
AATTGCCGGATTGGACTCTTTGGTTCAGATTAAAGAACNGTGACAATTCG
AGTCCGAACTGTAGCATGGGGCCACGGGCGCGTCTACACACCAGTTGA
AGGTAAAGACAGAAGATGTTGAGAAAATTGCATTCAGAGCGAGGTATGGT
CATTATGAATTCCTTGTTATGCCATTGGAGTAACTAATGCCCTGCAACGTT
TATAGACCTAATGAACGTATTTTAAAGACTTATCTTGATGATTTTGTGTGA
TTTTATTAATGATATCTTGGTGTATTCAAAGAATAGACTTGAACATGAACACC
ACTTGAGAACTTCTTGCAAAACACTTAAAGAAAAGAAATTATTTCCAAAC
TAAAAAATGTGAATTTGGTTGGATGAAGTCATTTCTTAGGGCATGTTAT
CAACAAGGAAGAAATTCAGTAGATCCACAGAAAATTGAAGCAATTGTGA
ATTGACCCACTCCAACAAATATAACAGAAGTCCATAGTTTCGTGGGCTIAGC
TGGGTACTACAGAAGATTGTGAAGG

upstream primer

(as shown in SEQ ID NO:8)

downstream primer

(as shown in SEQ ID NO:21)

FIG. 4

Scaffold3614:66549 (as shown in SEQ ID NO:10)

GAGTCATGGGTTTCTTAAATTTCTCTAAAAATATTAGGTGGTGACTCTGT
ATCTGGCAAAATAGTCCATTTTGGCAATTGATTCAAAATCAGTTTTCCAA
CATATTGCCGAATTGGGACTTTTGGTGATTATCTATTTACATTGCACATG
TGAAATCAGATTGAGAACCGTGGGAGTCCGATACTGTAGGGCTTATTCGTC
TTCCGAAAAGGGGCATGCAAAGTCGAACCTACAAGTCCCTGGGGAGGATG
GATTGCAAAATTACCGTACACAGTAGCAATCCCGTCTTTAAAGGOGTACTTT
ACCAACTGATGGACCATTGATGACACAACCCCTCATCTGATGTAGCCAGGG
TCTTCCAGTAGTAGATTGAAAGTGTCGAAAAATCCATGACATAGAATTTA
ACCTGATGCTCAGACGGGCCGAGTAGGATATGGCTCTTAAACATTACCATG
ACATCTTGGCTCGTATTGTCAATATAAGCCTAAACGGCCTGGGTCGTGGGCGT
AAAGTTAGTCGGCCTCACACCGATGGCATAGGCGTCCTTACCGGGCATA
CATTAATCGCCGATCCGTTATCTACCAACACCACTGGAATCCACTTTTCTG
ACTTCCAGCGTTACATATAAGGGCCAATTGTGGTTAGCACCOCTCAGGTGGT
AACTCTTTATCTATAAAAGATATCACTGGCGTAACATCCCCGGATGTAAACA
ATGATACCAATTGGTCAGCAGTGGTTTCGATAGGGAGTTTGGTCCGGTTCAT
TGCCCTAGCAGCAGTGCCTGTCTATGCTCCCGAGATGCCATGATTAGCCCC
CAGATTGATATGTCGGCCTGAATCTTCTTAAGCTGTTTCAAGACCAGGTTTT
CTTCAACATCCTTCTCTTTGATTTCTCGACCCCCACTGTCCTTGATATATGC
CATCTTTTAGGGTTATCACCCATTGGTACCCCTTTCGGTCTAGATTACCCTGA
CTTAAAGGTCTCCTTCTC/

upstream primer
(as shown in SEQ ID NO:11)

downstream primer
(as shown in SEQ ID NO:22)

FIG. 5

Scaffold349:3413816 (as shown in SEQ ID NO:13)

ATAATCTTTTGTACTTGTTCAGGTGGAATGAAGCAATCAACCGAGAGTCC
AGGAACATTGAATGCTAGGTCGTCGATCTTCCAAGTCTCCTCCATCCGTGT
GATTGCTGTGCCCCGCTCTCAGATTGTCCCCAAATCTTGAGATGATCACT
TGATTGGCCAGAAATGCGCGATCATCGTGCCCTCCACCATCCGATAGTCCTC
GATTTTCGTGCCCATGGTGGTCTCCCAATAGGTAGGTTAGGTTCGGGGG
ACTGGATTCTGGTGAGGTAAGAGTCTCTAAATACACTAGCAGACCACTT
CTTGGCTGAAGTAACCAAACATGACATGCTTGATCATCCTGCACTGT

upstream primer

GTCACCTCCGATCGGCTAGGTCCGTCTGATCCGCGGACAATTTCAACACGA (as shown in SEQ ID NO:14)

AGCAATCGACGCTCAAGATTCGTTTTTCGCCCACGTATTGTGCTAGGGAAA
ACACAGCCGATACAGCCACAGGATCTAGTCCCTGCATAAATAACANATG
TTTTTIACATAGAGGAAAAATAATATCTGTCACATGAATTCCTACTCCATTTT
TTAACCTTCTAAGAAAGTGTGGTGAAAAAAATATTAAATCCATTGGGTA

downstream primer

AAATATAACAGTCTTTAACATAACAATATGGCGAACTATACATTCAATTCT (as shown in SEQ ID NO:23)

AGAAAATGTCTCATTTTTATAGATTTTTATGAAAGGGATCAACCTTCTTTT
TTTTTATTGGAAGCACTATATAAATAATGTCAAATAGTTTTCCAACTTAT
CTAAATAAAGTTTTAATAATTTAATCCACACATTTGAATTTAATTACTT
ATTTTATAGTAGATAACATTACCACAGTCAAAAAGAGTGCCAACATGAACC
TCCAGCACACTTGAAGAGCACTTGACGATCATATTGGGAAAGTTACCAGC
CAGCACTCCCCAAAAAAGAAAAAAGATAAAAGATTAAAAAAA
TTAGTAAAAAGTGACTTTACAAAAAGGAATATTCCACCTCTG

FIG. 6

Scaffold920:281727 (as shown in SEQ ID NO:16)

AGGGAGACTTTTATCTTGAGAGCTAGAAGAAGAGAAAGTTAGAGAAAAGA
AAGAGAAGTAGGAAGAAAATCAAAGGGAATTCACATTCGTCCTTTTGGAG
TTGAGAATTGAACACTTAGGTGATTTGAAAATCATAAATGAGGTGTGTTA
AACTAATATCGTTCAGCTACAGTTACTCAGTAAATTCTCTTCTCAGAGGCT
ACGCAGGTGTAGTTTGAGTTAAACTTGGCCACTTAACTAATGGAACCATT
AGGGGCCCAAGCTAATTAGTTCCTAGAACAAAGGAGAGAGGACGGAGAA
GCATAGAGAAAGTTAGAGAGAACTTTTTTCTTGAGAGATAGAAGAGATAG
TTAGAGAAAAGAAAGAGAAACGGGAAAAAATCATTGGGAATTCGCATT
CGTCTTTTGGCCTTGAGAATTGAACAGTTGGGGAATTGGGAAACCTTA
AATGCGGTGCTTATGTTTAACTAATATCGTTAAGTGCCAATTACTCANTAAAT
CCTCTTCTTAGATGCTAAGCAAGATTAGTGTAGTTAACTTGGCCACTTA
AGCTAATGGAACAGTTAGGGTCCCAAGCGAATTAGTTTCCTAGAACAAAAG
ATAGAAGGATGGAGAATGTAGCACGTTTCGTGAGGGACCCGCTACTACA
GTTCGGACTCGATTGTGTACGGTTCTTAATCTGAACCAAAGAGTCCAAA
TCCGGCAAATCGTTTTGAGAAACAGATTTTTTGAAAAGAAGTGCCAAACAT
GGACTGCTTTGCTAGATATAGAGTCGCCACCTAAATATTTTTTAAAATGGG
GAAATTAGGAAACCTAACTTGGTGCCAAAAGCCACGTGTCCGTCATTGC
CAAAGTTGCCTGGGCTCGGGAGCTTGGGTACGATTGGGAAGGTCAGCTAT
GAGCACCCCTCTCGCCCGATCCGAAGATCGGCCTCTACTAACCGTGATATC
CGTTTTTGAAAACGTTATGTGTTCTTAAACCAATT

upstream primer
(as shown in SEQ ID NO:17)

downstream primer
(as shown in SEQ ID NO:24)

FIG. 7

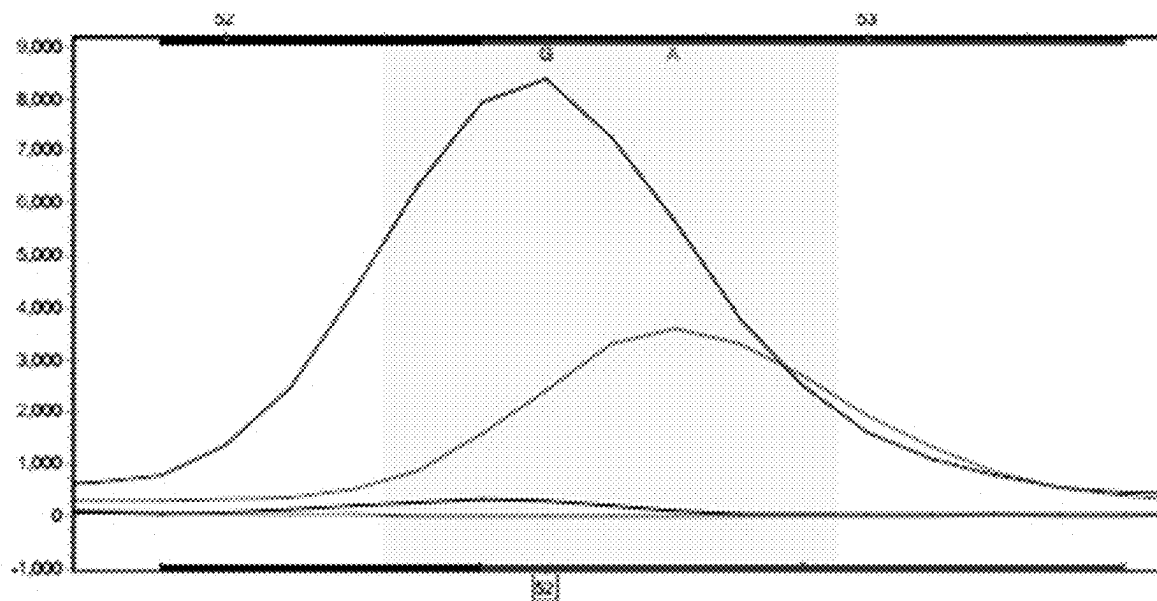


FIG. 8

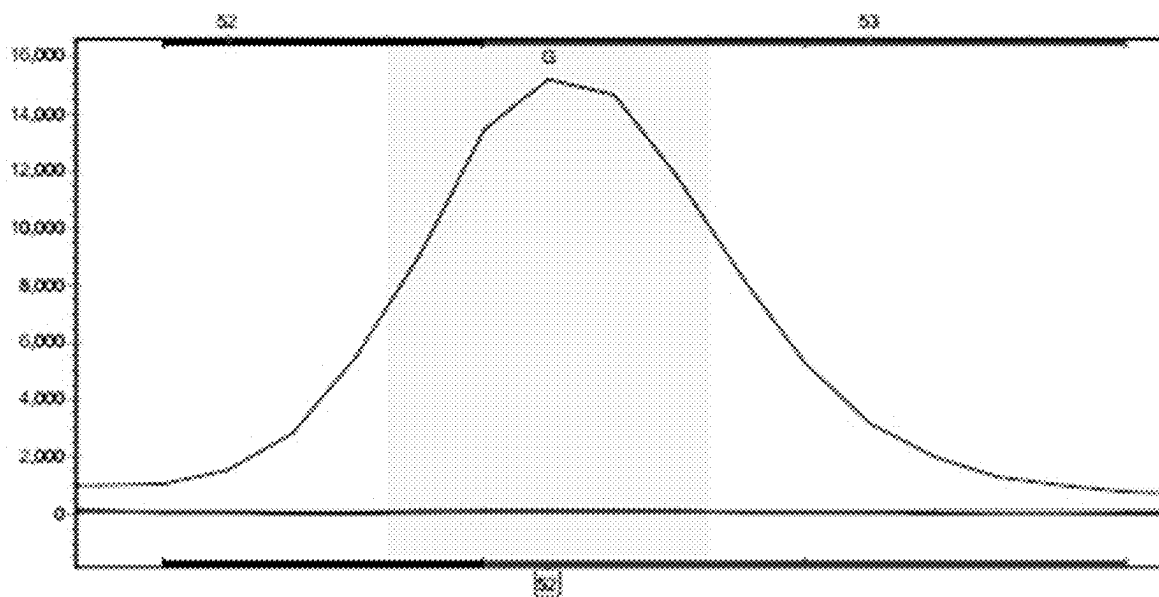


FIG. 9

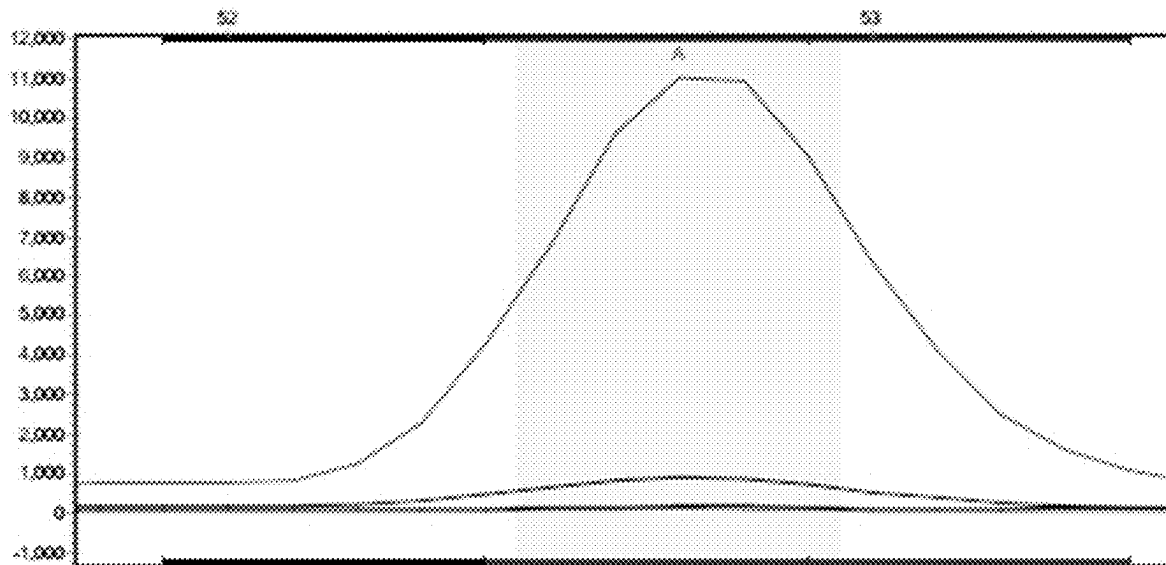


FIG. 10

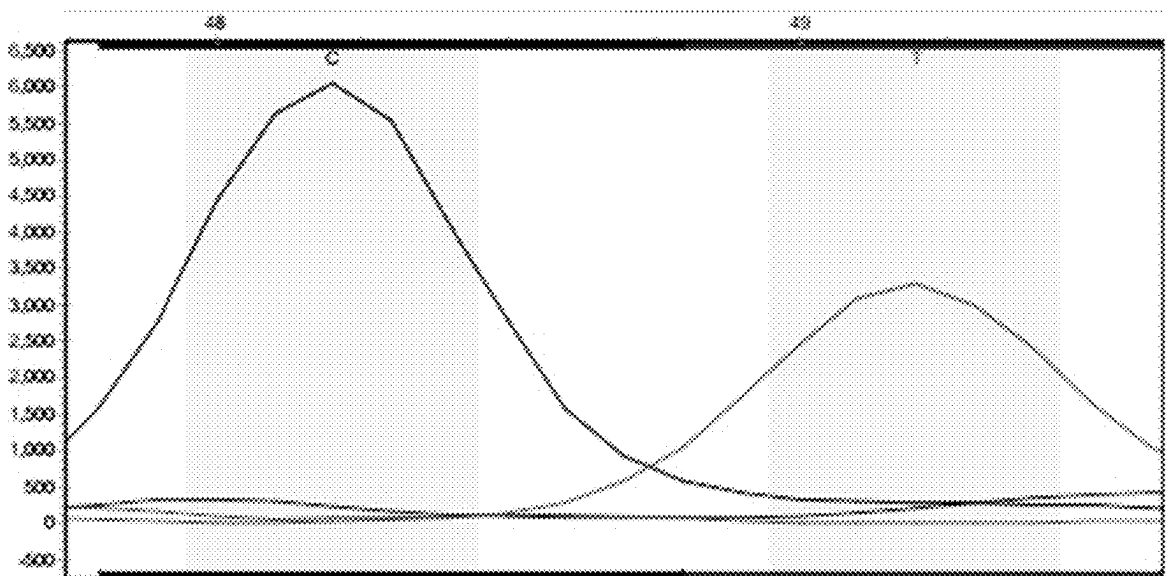


FIG. 11

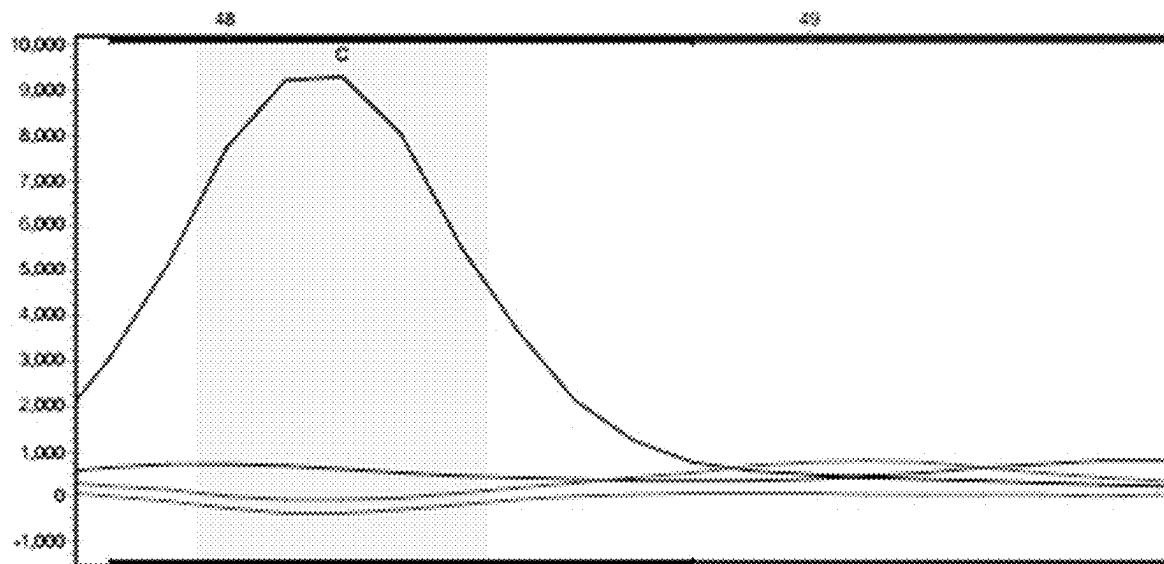


FIG. 12

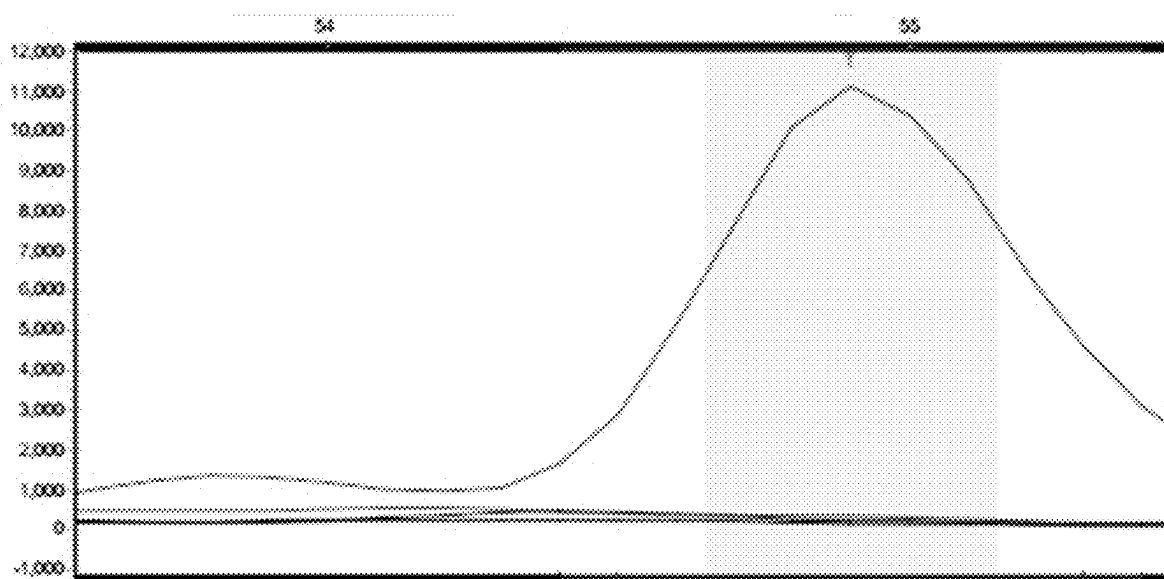


FIG. 13

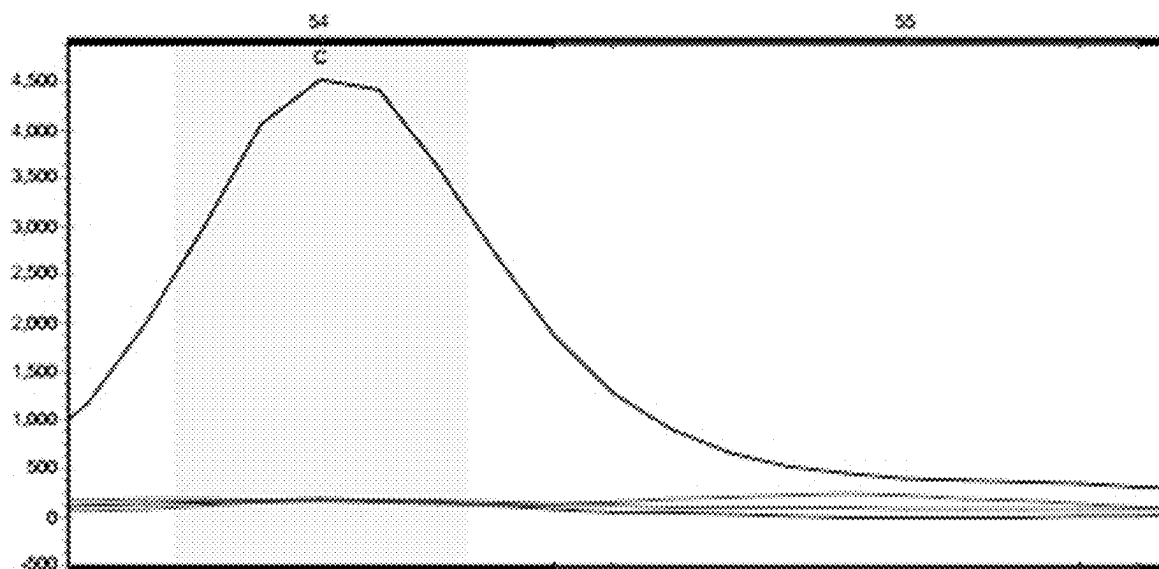


FIG. 14

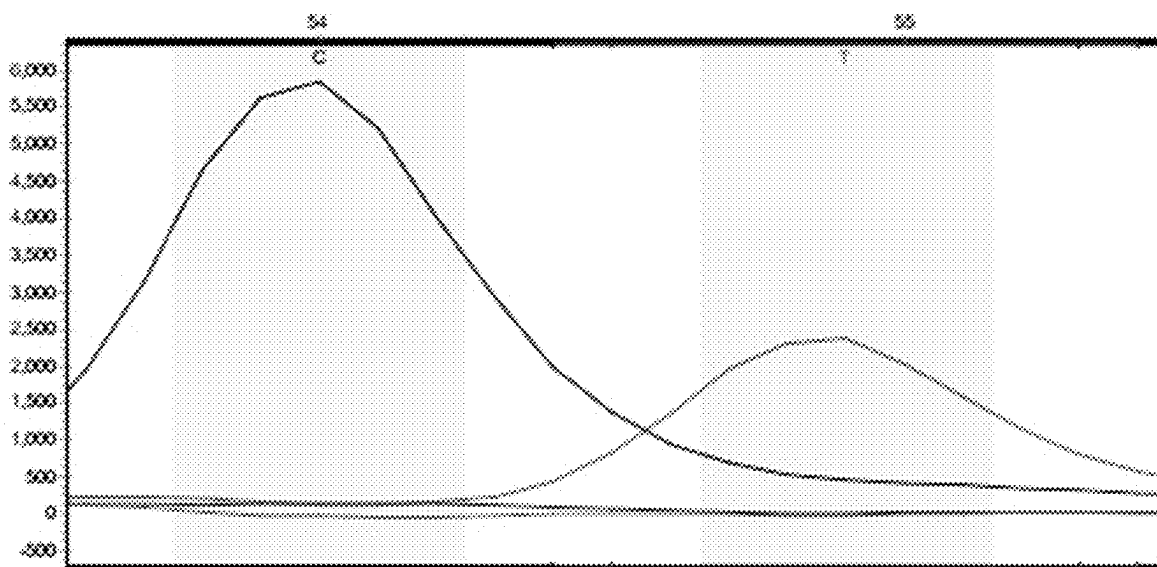


FIG. 15

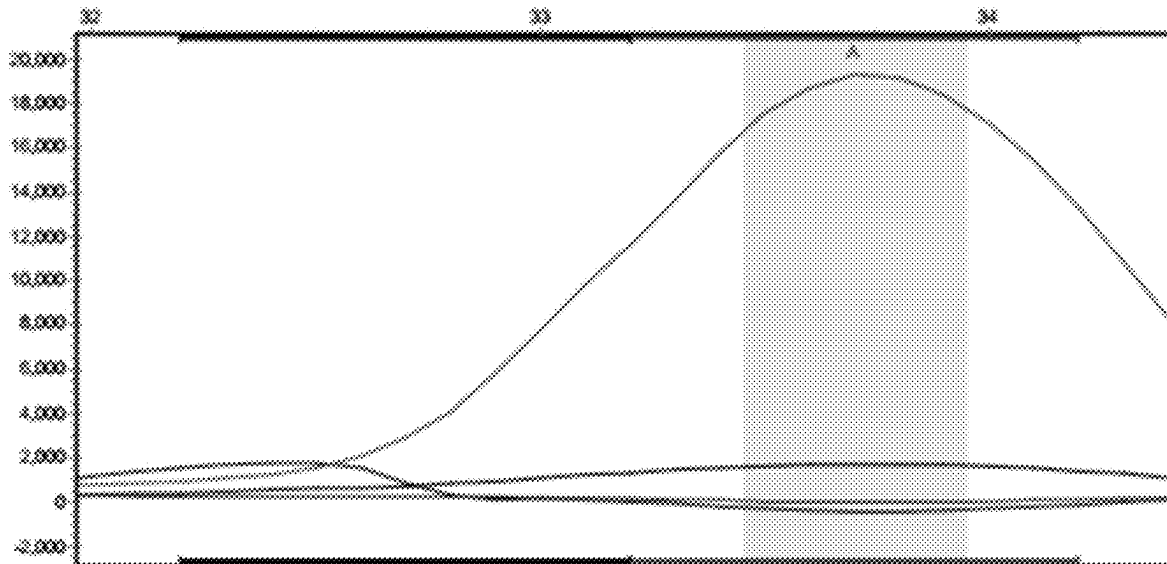


FIG. 16

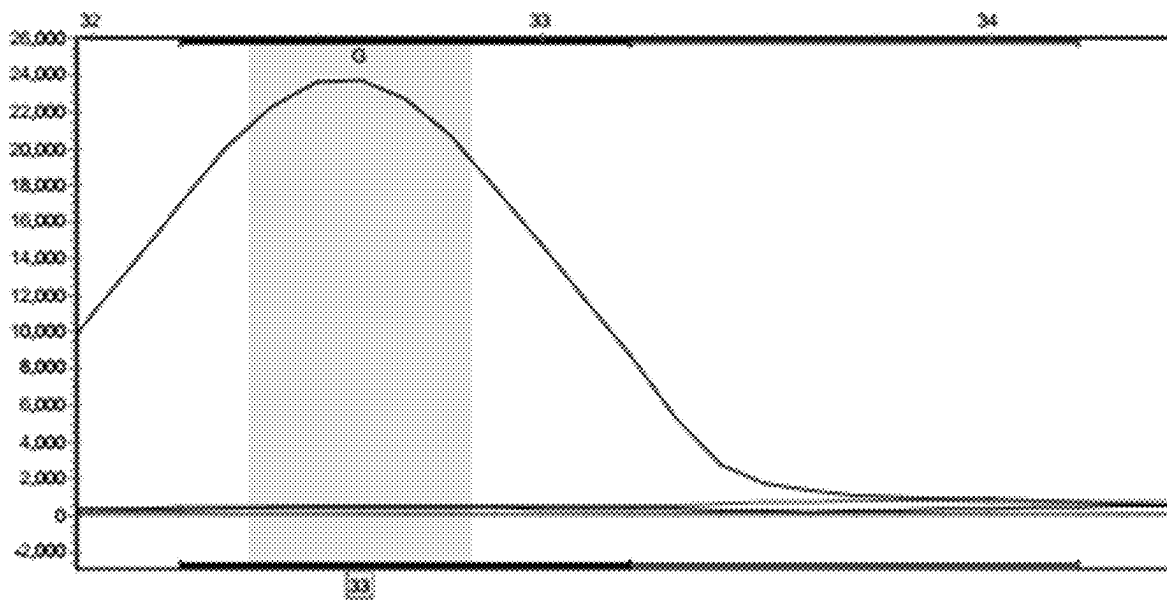


FIG. 17

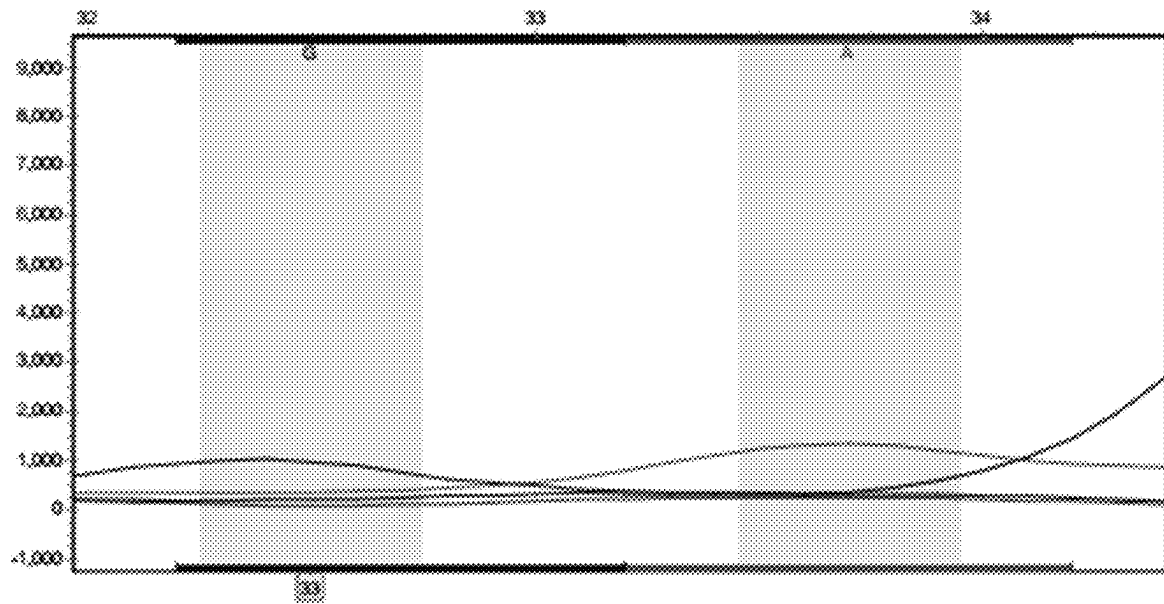


FIG. 18

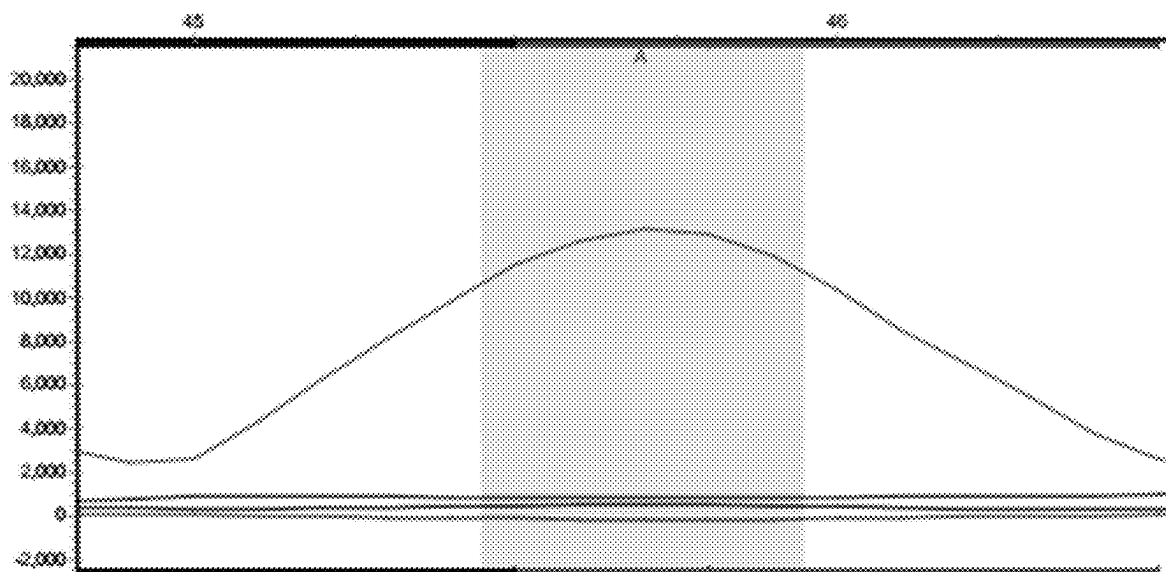


FIG. 19

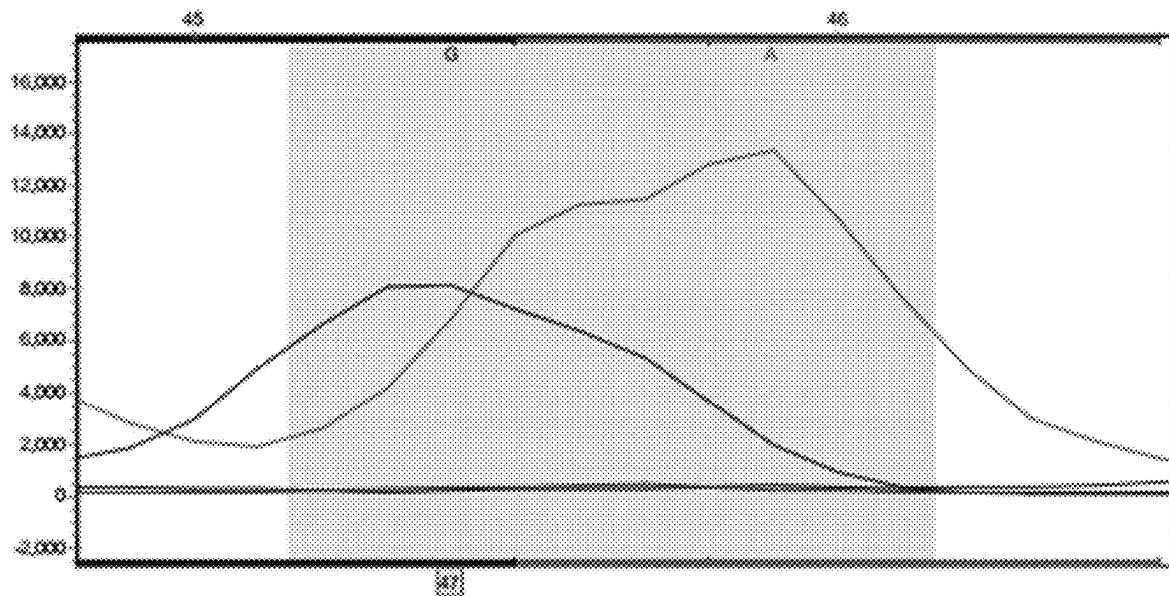


FIG. 20

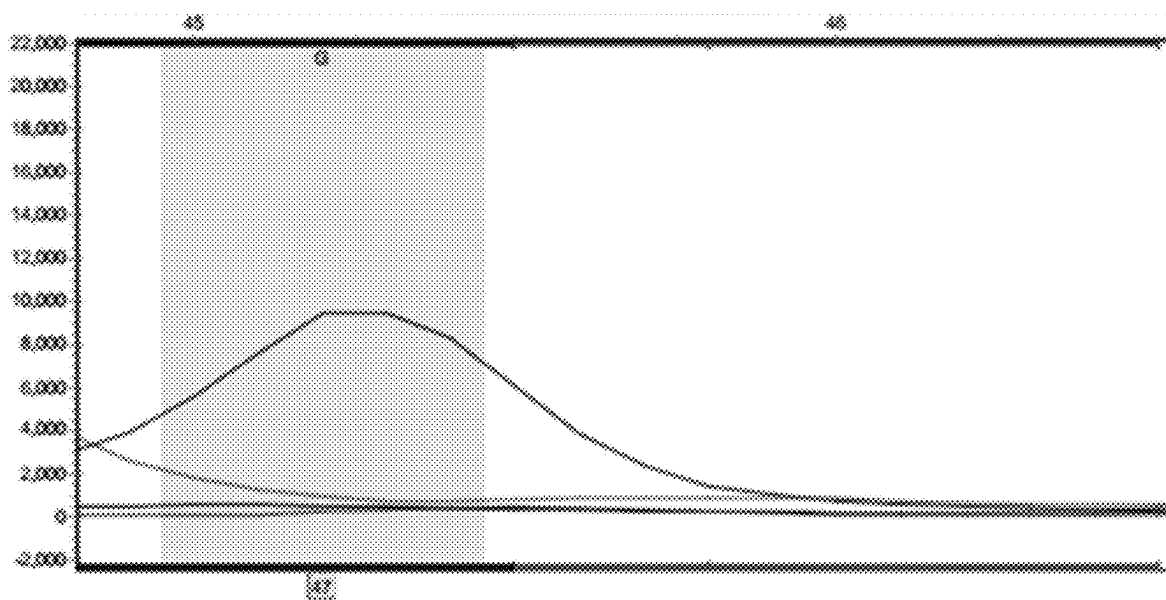


FIG. 21

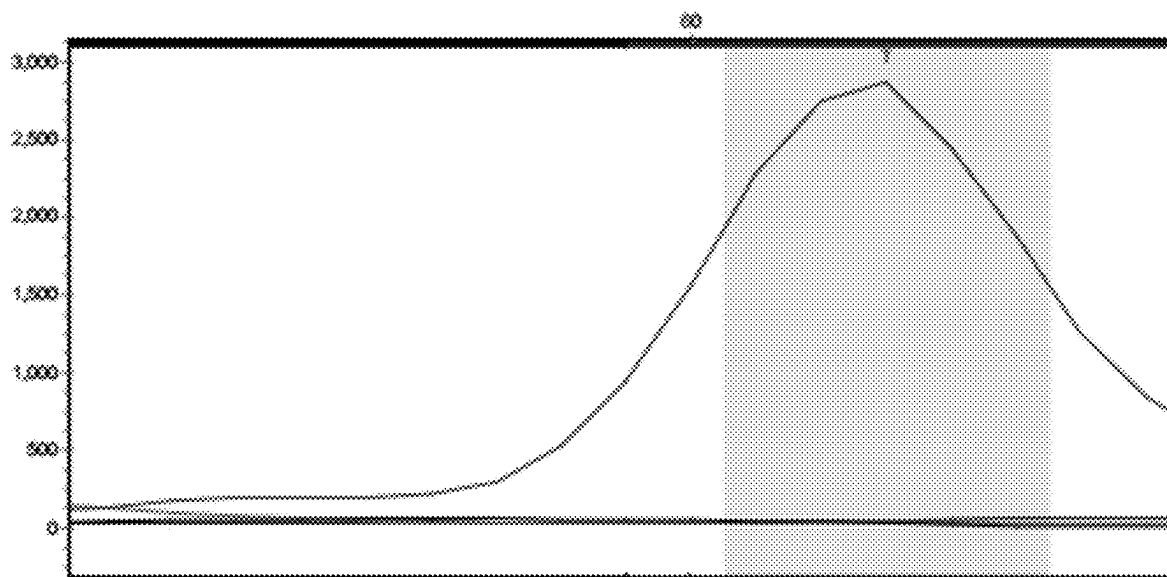


FIG. 22

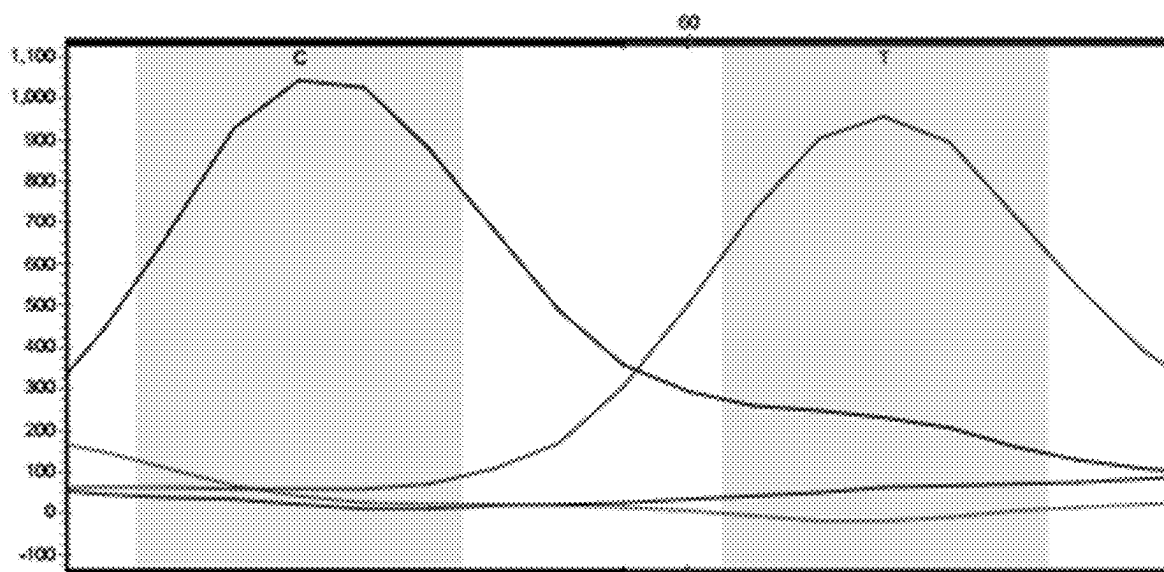


FIG. 23

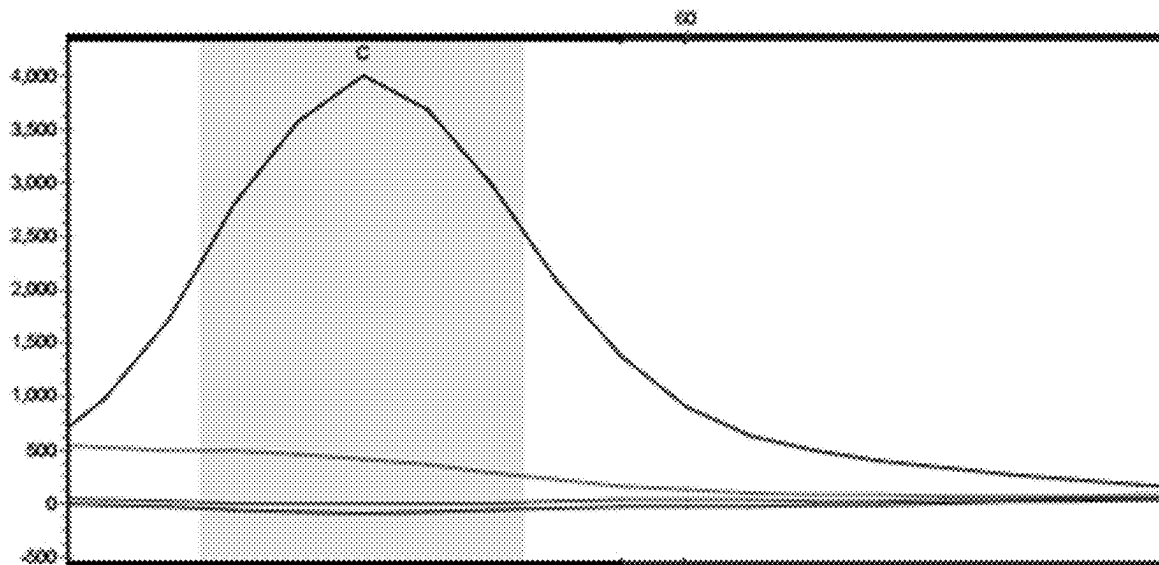


FIG. 24

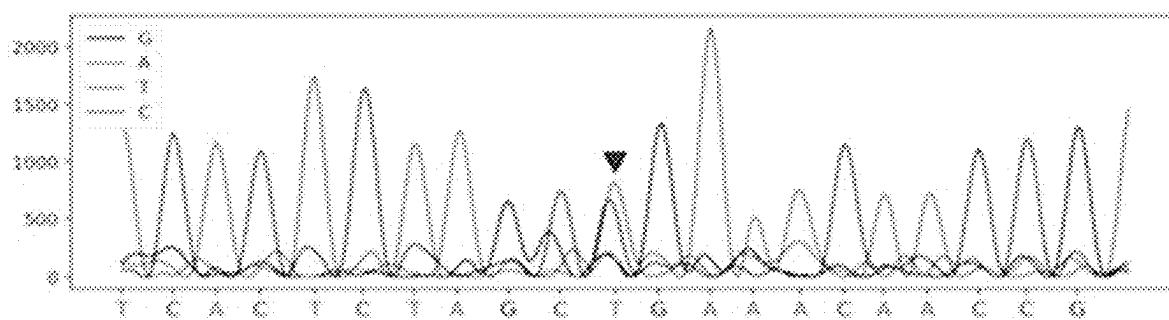


FIG. 25

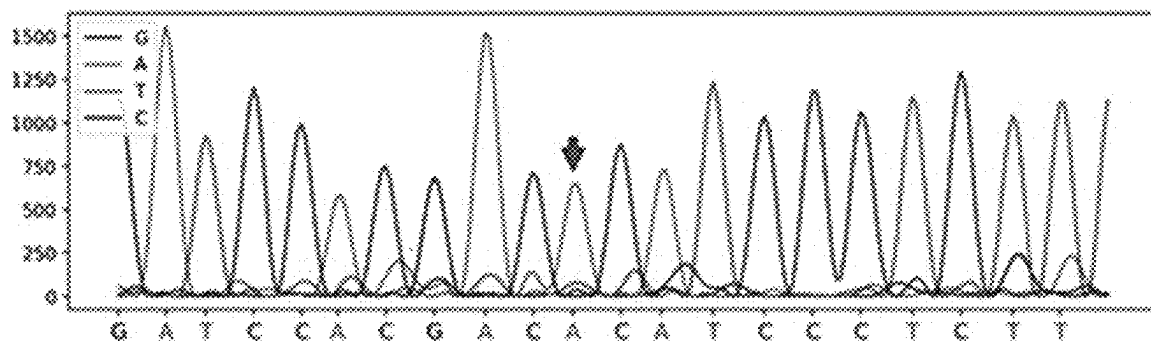


FIG. 26

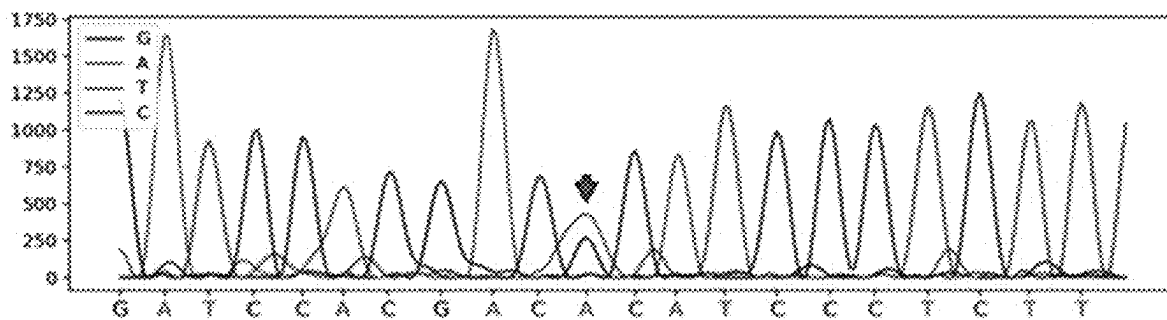


FIG. 27

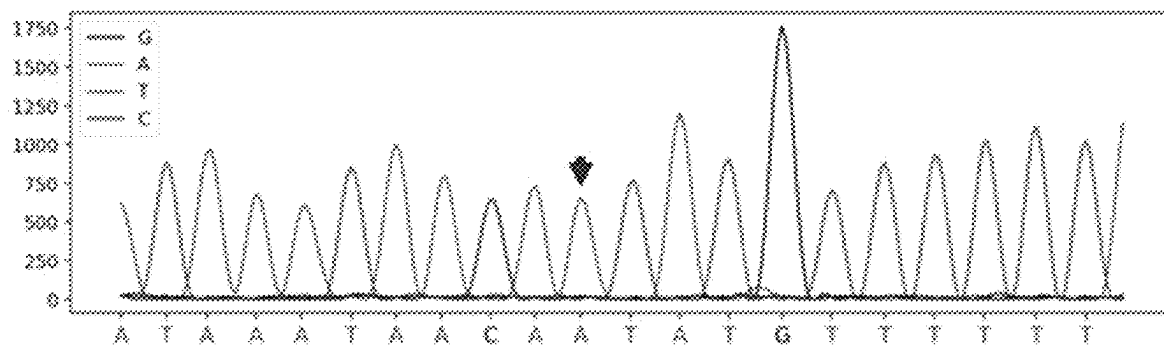


FIG. 28

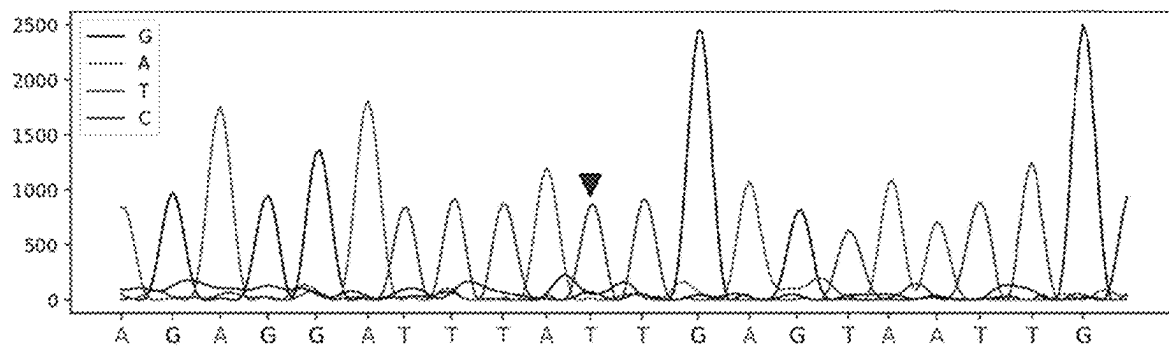


FIG. 29

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MOLECULAR MARKER COMBINATION LINKED TO QUANTITATIVE TRAITS OF TEA PLANT CAFFEINE CONTENT

CROSS-REFERENCE TO RELATED APPLICATION

This application is a 371 of international application of PCT application serial no. PCT/CN2019/110919, filed on Oct. 14, 2019, which claims the priority benefit of China application no. 201910833687.X, China application no. 201910834185.9, and China application no. 201910833668.7, filed on Sep. 4, 2019. The entirety of each of the above mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

The present invention relates to the technical field of molecular genetics and breeding, and more specifically, to a molecular marker combination linked to quantitative traits of tea plant caffeine content.

BACKGROUND

Tea (*Camellia sinensis* (L.) O.Kuntze) belongs to the genus *Camellia* (Theaceae), which originated in southwest China, with a cultivation history of more than 5,000 years. Tea, coffee, and cocoa are collectively referred to as the world's three major non-alcoholic beverages, which have important economic value and have an important impact on society and culture.

Caffeine, which is a characteristic secondary metabolite in tea shoots, is one of the main factors affecting tea flavor. Caffeine is a high content of alkaloids in tea, generally 2% to 5%. Each cup of 150 ml tea soup contains about 40 mg of caffeine. Caffeine is a central nervous system stimulant, so it has a refreshing effect. In addition, caffeine also has the effect of enhancing physical strength, perseverance and endurance. Studies have shown that caffeine can enhance muscle energy, especially for upper limb muscles. Caffeine can also promote metabolism.

Ingesting 200 mg of caffeine, the metabolism rate will increase by 7% in the next 3 hours, and the rate of fat burning will be greatly improved. Caffeine can help relieve pain, because it can accelerate the onset of other pain medications. Caffeine has the effect of improving antioxidant, and caffeine can double the effectiveness of antioxidant phenolics.

However, some people are very sensitive to caffeine, even a small amount of caffeine intake can cause severe insomnia, rapid heartbeat, and increased blood pressure, and the risk of myocardial infarction after intake of excessive caffeine is higher.

Existing research shows that caffeine varies from different producing area and different varieties. In order to meet the needs of different populations, tea plants with different caffeine content need to be cultivated. Meanwhile, due to the importance of caffeine to tea quality and physiological functions, it is of great significance to breed tea plant varieties with specific caffeine content. At present, tea plant breeding is mainly adopt conventional methods, that is, excellent individual plants are selected from wild populations and hybrid offspring for systematic breeding. This method is time-consuming and inefficient, which makes the replacement of new varieties slow, and it cannot quickly

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meet the public's demand for new products. Since molecular marker-assisted breeding can select breeding materials at the seedling stage, it can significantly improve breeding efficiency. The discovery of molecular markers that are closely linked to the excellent traits of the tea plant is the basis for the molecular marker-assisted selective breeding of tea plant.

SUMMARY OF THE INVENTION

Objectives of the present invention are to overcome the shortcomings of the prior art and provide a molecular marker combination linked to quantitative traits of tea plant caffeine content.

The first objective of the present invention is to provide a molecular marker combination linked to quantitative traits of tea plant caffeine content. The molecular marker combination comprises a SNP site 1, a SNP site 2, a SNP site 3, a SNP site 4, a SNP site 5 and a SNP site 6, which are located in tea genomes Scaffold4239:309117, Scaffold115:803980, Scaffold720:596655, Scaffold3614:66549, Scaffold349:3413816 and Scaffold920:281727, respectively, which are a 501st base of a nucleotide sequence shown in SEQ ID NO: 1, a 501st base of a nucleotide sequence shown in SEQ ID NO: 4, a 501st base of a nucleotide sequence shown in SEQ ID NO: 7, a 501st base of a nucleotide sequence shown in SEQ ID NO: 10, a 501st base of a nucleotide sequence shown in SEQ ID NO: 13, and a 501st base of a nucleotide sequence shown in SEQ ID NO: 16.

The second objective of the present invention is to provide use of any one or more molecular markers of the molecular marker combination in evaluating the tea plant caffeine content.

The third objective of the present invention is to provide use of primers of any one or more molecular markers of the molecular marker combination in evaluating the tea plant caffeine content.

The fourth objective of the present invention is to provide primers for detecting SNP site 1.

The fifth objective of the present invention is to provide primers for detecting SNP site 2.

The sixth objective of the present invention is to provide primers for detecting SNP site 3.

The seventh objective of the present invention is to provide primers for detecting SNP site 4.

The eighth objective of the present invention is to provide primers for detecting SNP site 5.

The ninth objective of the present invention is to provide primers for detecting SNP site 6.

The tenth objective of the present invention is to provide a kit for evaluating tea plant caffeine content.

The eleventh objective of the present invention is to provide a method for evaluating tea plant caffeine content.

The twelfth objective of the present invention is to provide use of one or more of any one or more molecular markers of the molecular marker combination, the primers for the SNP site 1, the primers for the SNP site 2, the primers for the SNP site 3, the primers for the SNP site 4, the primers for the SNP site 5, the primers for the SNP site 6, or the kit in molecular-assisted breeding.

In order to achieve the above objectives, the present invention is realized by the following technical solutions.

After a long period of exploratory research, the inventors discovered six SNP site molecular markers linked to caffeine content. It is further used to establish a detection method for

detecting the sites, which can be used to evaluate the tea plant caffeine content, for further use in resource screening and molecular breeding.

Therefore, the present invention claims a molecular marker combination linked to quantitative traits of tea plant caffeine content, including a SNP site 1, a SNP site 2, a SNP site 3, a SNP site 4, a SNP site 5 and a SNP site 6, which are located in tea genomes Scaffold4239: 309117, Scaffold115: 803980, Scaffold720:596655, Scaffold3614:66549, Scaffold349:3413816 and Scaffold920:281727, respectively, i.e., a 501st base of a nucleotide sequence shown in SEQ ID NO: 1, a 501st base of a nucleotide sequence shown in SEQ ID NO: 4, a 501st base of a nucleotide sequence shown in SEQ ID NO: 7, a 501st base of a nucleotide sequence shown in SEQ ID NO: 10, a 501st base of a nucleotide sequence shown in SEQ ID NO: 13, and a 501st base of a nucleotide sequence shown in SEQ ID NO: 16.

The SNP site 1 is located in the tea genome Scaffold4239: 309117 (i.e. the 501st base of the nucleotide sequence shown in SEQ ID NO: 1), this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the tea plant caffeine content in the dry matter of tea soup corresponding to an AA genotype sample has extremely significant difference compared with GG and GA genotype samples. It is statistically judged that, when the genotype of the sample is double mutant AA, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type GG or single mutant GA.

The SNP site 2 is located in the tea genome Scaffold115: 803980 (i.e. the 501st base of the nucleotide sequence shown in SEQ ID NO: 4), this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of the tea plant corresponding to a GG genotype sample has extremely significant difference compared with AA and GA genotype samples. It is statistically judged that, when the genotype of the sample is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the sample of which the genotype is wild type AA or single mutant GA.

The SNP site 3 is located in the tea genome Scaffold720: 596655 (i.e. the 501st base of the nucleotide sequence shown in SEQ ID NO: 7), this site is T or C, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of the tea plant corresponding to a CC genotype sample has extremely significant difference compared with a CT genotype sample. It is statistically judged that, when the genotype of the sample is single mutant CT, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the CC type.

The SNP site 4 is located in the tea genome Scaffold3614: 66549 (i.e. the 501st base of the nucleotide sequence shown in SEQ ID NO: 10), this site is C or T, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of tea soup corresponding to a CC genotype sample has extremely significant difference compared with TT and CT genotype

samples. It is statistically judged that, when the genotype is double mutant CC, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type TT or single mutant CT.

The SNP site 5 is located in the tea genome Scaffold349: 3413816 (i.e. the 501st base of the nucleotide sequence shown in SEQ ID NO: 13), this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of tea soup corresponding to a GG genotype sample has extremely significant difference compared with GA and AA genotype samples. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

The SNP site 6 is located in the tea genome Scaffold920: 281727 (i.e. the 501st base of the nucleotide sequence shown in SEQ ID NO: 16), this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of tea soup corresponding to a GG genotype sample has extremely significant difference compared with GA and AA genotype samples. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

The tea plant caffeine content according to the present invention is specifically a proportion of caffeine in dry matter of fresh tea leaves.

Use of any one or more molecular markers of the molecular marker combination in evaluating the tea plant caffeine content also belongs to the scope of protection of the present invention.

The present invention further claims use of primers of any one or more molecular markers of the molecular marker combination in evaluating the tea plant caffeine content.

Primers for the SNP site 1, wherein nucleotide sequences thereof are shown as SEQ ID NO: 2 and SEQ ID NO: 3.

```

primer F: (SEQ ID NO: 2)
GAAGACTAACCCGTATCGAG;

primer R: (SEQ ID NO: 3)
ACACTTACAGTCTCTTGCGG.

```

Primers for the SNP site 2, wherein nucleotide sequences thereof are shown as SEQ ID NO: 5 and SEQ ID NO: 6.

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primer F: (SEQ ID NO: 5)
CTTCATCTCCACCACACTTC;

primer R: (SEQ ID NO: 6)
GCCCAAAGTAGCAAAGAGAG.

```

Primers for the SNP site 3, wherein nucleotide sequences thereof are shown as SEQ ID NO: 8 and SEQ ID NO: 9.

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primer F: (SEQ ID NO: 8)
CAACTTTGGTGATGACGGAC;
primer R: (SEQ ID NO: 9)
TTCAACTGGTGTGTAGACGC.

Primers for the SNP site 4, wherein nucleotide sequences thereof are shown as SEQ ID NO: 11 and SEQ ID NO: 12.

primer F: (SEQ ID NO: 11)
GATGACACAACCCTCATCTG;
primer R: (SEQ ID NO: 12)
AATGTATGCCCGTAAGGAC.

Primers for the SNP site 5, wherein nucleotide sequences thereof are shown as SEQ ID NO: 14 and SEQ ID NO: 15.

primer F: (SEQ ID NO: 14)
TCTCTGCACTGTTGTCACTC;
primer R: (SEQ ID NO: 15)
CACCACACTTCTCTAGAAGG.

Primers for the SNP site 6, wherein nucleotide sequences thereof are shown as SEQ ID NO: 17 and SEQ ID NO: 18.

primer F: (SEQ ID NO: 17)
TTCGCATTCGTCCTTTTGGG;
primer R: (SEQ ID NO: 18)
ACGTGCTACATTCTCCATCC.

Further, the present invention claims a kit for evaluating tea plant caffeine content, including a reagent for detecting the molecular marker combination or any one molecular marker thereof.

Preferably, the reagent is the primers for the SNP site 1 which have the nucleotide sequences shown as SEQ ID NO: 2 and SEQ ID NO: 3, the primers for the SNP site 2 which have the nucleotide sequences shown as SEQ ID NO: 5 and SEQ ID NO: 6, the primers for the SNP site 3 which have the nucleotide sequences shown as SEQ ID NO: 8 and SEQ ID NO: 9, the primers for the SNP site 4 which have the nucleotide sequences shown as SEQ ID NO: 11 and SEQ ID NO: 12, the primers for the SNP site 5 which have the nucleotide sequences shown as SEQ ID NO: 14 and SEQ ID NO: 15, and/or the primers for SNP site 6 which have the nucleotide sequences shown as SEQ ID NO: 17 and SEQ ID NO: 18.

The most preferably, the kit contains the primers for the SNP site 1 which have the nucleotide sequences shown as SEQ ID NO: 2 and SEQ ID NO: 3, the primers for the SNP site 2 which have the nucleotide sequences shown as SEQ ID NO: 5 and SEQ ID NO: 6, the primers for the SNP site 3 which have the nucleotide sequences shown as SEQ ID NO: 8 and SEQ ID NO: 9, the primers for the SNP site 4 which have the nucleotide sequences shown as SEQ ID NO: 11 and SEQ ID NO: 12, the primers for the SNP site 5 which have the nucleotide sequences shown as SEQ ID NO: 14 and SEQ ID NO: 15, the primers for the SNP site 6 which have

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the nucleotide sequences shown as SEQ ID NO: 17 and SEQ ID NO: 18, 2×Taq PCR Master Mix, and ddH₂O.

A usage method is as follows:

- (1) CTAB method is used to extract total DNA from buds of tea plant, it is ensured that A260/A280 of each DNA sample is between 1.8 and 2.0, and the concentration is greater than 100 µg/µl;
- (2) PCR Amplification
PCR system (10 µl) is as follows:

2× Taq PCR Master Mix	5 µl
primer	Each 0.5 µl
DNA template	1 µl
ddH ₂ O	3 µl

PCR amplification procedure is as follows:

95° C.	5 minutes	
95° C.	30 seconds	×45 cycles
56° C.	30 seconds	
72° C.	30 seconds	
72° C.	2 minutes	
4° C.	forever	

(3) Product Purification

The PCR amplification products are subjected to gel electrophoresis, followed by recovery and purification using a commercially available gel electrophoresis DNA recovery kit.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 2 and SEQ ID NO: 3 is selected for recovery and purification.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 5 and SEQ ID NO: 6 is selected for recovery and purification.

A band with a fragment length of about 250 bp in the amplification product of the primers shown in SEQ ID NO: 8 and SEQ ID NO: 9 is selected for recovery and purification.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 11 and SEQ ID NO: 12 is selected for recovery and purification.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 14 and SEQ ID NO: 15 is selected for recovery and purification.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 17 and SEQ ID NO: 18 is selected for recovery and purification.

(4) Sequencing and Interpretation of Results

The recovered and purified product is sent to a sequencing company for Sanger sequencing. At the site Scaffold4239: 309117, it is statistically judged that, when the genotype sample is double mutant AA, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type GG or single mutant GA.

At the site Scaffold115:803980, it is statistically judged that, when the genotype of the sample is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

At the site Scaffold720:596655, it is statistically judged that, when the genotype is single mutant CT, the caffeine content in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is CC.

At the site Scaffold3614:66549, it is statistically judged that, when the genotype is double mutant CC, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type TT or single mutant CT.

At the site Scaffold349:3413816, it is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

At the site Scaffold920:281727, it is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

In the meantime, the present invention claims a method for evaluating the tea plant caffeine content, which detects a genotype of any one or more molecular markers of the molecular marker combination.

Use of any one or more of any one or more molecular markers of the molecular marker combination, the primers for the SNP site 1, the primers for the SNP site 2, the primers for the SNP site 3, the primers for the SNP site 4, the primers for the SNP site 5, the primers for the SNP site 6 or the kit in molecular-assisted breeding.

Compared with the prior art, the present invention has the following beneficial effects.

The present invention first discovered a molecular marker combination linked to quantitative traits of tea plant caffeine content, which includes a SNP site 1, a SNP site 2, a SNP site 3, a SNP site 4, a SNP site 5 and a SNP site 6, which are located in tea genomes Scaffold4239:309117, Scaffold115:803980, Scaffold720:596655, Scaffold3614:66549, Scaffold349:3413816 and Scaffold920:281727, and genotypes thereof are all extremely significantly correlated with the caffeine content.

The SNP site 1 is located in the tea genome Scaffold4239:309117, this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the tea plant caffeine content in the dry matter of tea soup corresponding to an AA genotype sample has extremely significant difference compared with GG and GA genotype samples. It is statistically judged that, when the genotype sample is double mutant AA, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type GG or single mutant GA.

The SNP site 2 is located in the tea genome Scaffold115:803980, this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of the tea plant corresponding to a GG genotype sample has extremely significant difference compared with AA and GA genotype samples. It is statistically judged that, when the genotype of the sample is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than genotype is wild type AA or single mutant GA.

The SNP site 3 is located in the tea genome Scaffold720:596655, this site is T or C, and genotype thereof is extremely

significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of the tea plant corresponding to a CC genotype sample has extremely significant difference compared with a CT genotype sample. It is statistically judged that, when the genotype of the sample is single mutant CT, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the CC type.

The SNP site 4 is located in the tea genome Scaffold3614:66549, this site is C or T, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of tea soup corresponding to a CC genotype sample has extremely significant difference compared with TT and CT genotype samples. It is statistically judged that, when the genotype is double mutant CC, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type TT or single mutant CT.

The SNP site 5 is located in the tea genome Scaffold349:3413816, this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of tea soup corresponding to a GG genotype sample has extremely significant difference compared with GA and AA genotype samples. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

The SNP site 6 is located in the tea genome Scaffold920:281727, this site is G or A, and genotype thereof is extremely significantly correlated with the caffeine content in the dry matter of the tea plant. It is shown by correlation analysis and significance analysis verification that the caffeine content in the dry matter of tea soup corresponding to a GG genotype sample has extremely significant difference compared with GA and AA genotype samples. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

It is further established a detection method for detecting the six SNP sites, which can be used to evaluate the caffeine content of the tea plant, for further use in screening of tea plant resources and molecular breeding. This is the basis for molecular marker-assisted selective breeding for tea plant, which has great research value.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows caffeine content in different seasons.

FIG. 2 shows a schematic diagram of a site Scaffold4239:309117 (as shown in SEQ ID NO: 1) and primers, wherein N denotes a base to be tested at Scaffold4239:309117, and bold and underlined parts denote upstream and downstream primers (upstream primer as shown in SEQ ID NO: 2, downstream primer: CCGCAAGAGACTGTAAAGTGT (SEQ ID NO: 19)).

FIG. 3 shows a schematic diagram of a site Scaffold115:803980 (as shown in SEQ ID NO: 4) and primers, wherein N denotes a base to be tested at Scaffold115:803980, and bold and underlined parts denote upstream and downstream

primers (upstream primer as shown in SEQ ID NO: 5, downstream primer: CTCTCTTTGCTACTTTGGGC (SEQ ID NO: 20)).

FIG. 4 shows a schematic diagram of a site Scaffold720:596655 (as shown in SEQ ID NO: 7) and primers, wherein N denotes a base to be tested at Scaffold720:596655, and bold and underlined parts denote upstream and downstream primers (upstream primer as shown in SEQ ID NO: 8, downstream primer: GCGTCTACACACCAGTTGAA (SEQ ID NO: 21)).

FIG. 5 shows a schematic diagram of a site Scaffold3614:66549 (as shown in SEQ ID NO: 10) and primers, wherein N denotes a base to be tested at Scaffold3614:66549, and bold and underlined parts denote upstream and downstream primers (upstream primer as shown in SEQ ID NO: 11, downstream primer: GTCCTTACCGGCATACATT (SEQ ID NO: 22)).

FIG. 6 shows a schematic diagram of a site Scaffold349:3413816 (as shown in SEQ ID NO: 13) and primers, wherein N denotes a base to be tested at Scaffold349:3413816, and bold and underlined parts denote upstream and downstream primers (upstream primer as shown in SEQ ID NO: 14, downstream primer: CCTTCTAAGAAAGTGTGGTG (SEQ ID NO: 23)).

FIG. 7 shows a schematic diagram of a site Scaffold920:281727 (as shown in SEQ ID NO: 16) and primers, wherein N denotes a base to be tested at Scaffold920:281727, and bold and underlined parts denote upstream and downstream primers (upstream primer as shown in SEQ ID NO: 17, downstream primer: GGATGGAGAATGTAGCACGT (SEQ ID NO: 24)).

FIG. 8 shows SNaPshot sequencing results of genotype of the sample 2-72 at the site Scaffold4239:309117.

FIG. 9 shows SNaPshot sequencing results of genotype at the site Scaffold4239:309117 of the sample 2-78.

FIG. 10 shows SNaPshot sequencing results of genotype of the sample 2-97 at the site Scaffold4239:309117.

FIG. 11 shows SNaPshot sequencing results of genotype of the sample 2-77 at the site Scaffold720:596655.

FIG. 12 shows SNaPshot sequencing results of genotype of the sample 2-81 at the site Scaffold720:596655.

FIG. 13 shows SNaPshot sequencing results of genotype of the sample 2-23 at the site Scaffold115:803980 (reverse complement).

FIG. 14 shows SNaPshot sequencing results of genotype of the sample 2-97 at the site Scaffold115:803980 (reverse complement).

FIG. 15 shows SNaPshot sequencing results of genotype of the sample 2-80 at the site Scaffold115:803980 (reverse complement).

FIG. 16 shows SNaPshot sequencing results of genotype of the sample 2-70 at the site Scaffold3614:66549 (reverse complement).

FIG. 17 shows SNaPshot sequencing results of genotype of the sample 2-77 at the site Scaffold3614:66549 (reverse complement).

FIG. 18 shows SNaPshot sequencing results of genotype of the sample 2-72 at the site Scaffold3614:66549 (reverse complement).

FIG. 19 shows SNaPshot sequencing results of genotype of the sample 2-69 at the site Scaffold349:3413816.

FIG. 20 shows SNaPshot sequencing results of genotype of the sample 2-68 at the site Scaffold349:3413816.

FIG. 21 shows SNaPshot sequencing results of genotype of the sample 2-77 at the site Scaffold349:3413816.

FIG. 22 shows SNaPshot sequencing results of genotype of the sample 2-72 at the site Scaffold920:281727 (reverse complement).

FIG. 23 shows SNaPshot sequencing results of genotype of the sample 2-94 at the site Scaffold920:281727 (reverse complement).

FIG. 24 shows SNaPshot sequencing results of genotype of the sample 2-97 at the site Scaffold920:281727 (reverse complement).

FIG. 25 shows sequencing results of genotype at the site Scaffold4239:309117.

FIG. 26 shows sequencing results of genotype at the site Scaffold115:803980, AA genotype.

FIG. 27 shows sequencing results of genotype at the site Scaffold115:803980, GA genotype.

FIG. 28 shows sequencing results of genotype at the site Scaffold349:3413816.

FIG. 29 shows sequencing results of genotype at the site Scaffold920:281727.

DETAILED DESCRIPTION OF THE INVENTION

The present invention will be further described in detail below with reference to the accompanying drawings and specific embodiments, and the embodiments are only used to explain the present invention, and are not used to limit the scope of the present invention. The test methods used in the following embodiments are all conventional methods unless otherwise specified. The materials and agents used, unless otherwise specified, are the agents and materials available from commercial sources.

Embodiment 1

I. Experiment Sample

191 tea plant materials located in Guangdong Province Tea Plant Germplasm Resource Bank (Yingde, Guangdong, 113.30E, 24.30N) were collected, including 124 from Guangdong, 20 from Fujian, 14 from Guangxi, 9 from Zhejiang, 6 from Hunan, 6 from Yunnan, 1 from Jiangxi, 1 from Guizhou, 1 from Taiwan, and 8 offspring of Kenyan tea, 1 offspring of Georgian species. The selected materials are widely representative.

The selected resources are randomly distributed in the resource bank. Double row per plant was used, each row is 4 m, the row spacing is 1.5 m, and the plant spacing is 35 cm. The resource bank was subjected to conventional water and fertilizer management. At the end of 2016, the resources were pruned and deep pits were applied with base fertilizer, 4 tons of organic fertilizer, 0.75 tons of peanut bran and 5 kg of compound fertilizer per acre. After picking spring tea and summer tea in 2017, pruning and topdressing outside the root were conducted, 15 kg compound fertilizer and 30 kg urea per acre. On Mar. 15, 2017, Jun. 25, 2017, and Sep. 28, 2017, the new shoots (one bud with two leaves) of the tea plant were picked, to make steamed green samples, and tea soup was prepared according to water extraction method.

II. Phenotypic Data Analysis

1. Experimental Procedure

The high-performance liquid chromatography was used to detect caffeine in tea soup related to the taste of tea plant, referring to the Chinese standard detection method.

2. Experimental Results

Caffeine content is shown in Table 1.

TABLE 1

Percentage of caffeine in dry matter from different resources in different seasons:			
Sample	Caffeine content (%)		
	Spring	Summer	Autumn
Sample 1	1.98	1.42	2.23
Sample 2	2.88	2.05	2.64
Sample 3	2.16	2.31	2.16
Sample 4	2.49	2.28	3.07
Sample 5	2.49	2.24	2.90
Sample 6	1.98	1.75	2.58
Sample 7	1.73	2.01	2.58
Sample 8	2.00	2.36	3.12
Sample 9	3.75	2.35	3.42
Sample 10	2.49	2.36	2.82
Sample 11	3.17	2.66	3.31
Sample 12	1.85	3.12	2.56
Sample 13	2.46	2.53	2.60
Sample 14	1.61	2.54	3.72
Sample 15	3.94	2.36	2.15
Sample 16	3.27	2.75	3.30
Sample 17	2.52	1.79	2.59
Sample 18	2.80	1.92	3.02
Sample 19	3.40	1.98	2.93
Sample 20	2.81	2.04	2.22
Sample 21	2.14	1.86	1.98
Sample 22	3.39	2.67	3.12
Sample 23	4.39	2.70	1.91
Sample 24	2.70	2.76	2.55
Sample 25	3.06	2.40	1.91
Sample 26	2.48	3.41	2.98
Sample 27	2.29	1.56	2.77
Sample 28	2.47	2.16	2.46
Sample 29	0.09	0.09	0.11
Sample 30	2.63	2.69	3.19
Sample 31	2.77	1.79	3.27
Sample 32	2.67	2.31	3.36
Sample 33	2.24	3.36	2.52
Sample 34	2.51	2.43	2.62
Sample 35	2.27	2.34	2.47
Sample 36	2.73	2.98	2.82
Sample 37	1.91	2.45	3.37
Sample 38	2.74	1.81	3.02
Sample 39	3.54	2.20	3.15
Sample 40	2.43	2.55	2.46
Sample 41	3.10	2.28	2.46
Sample 42	2.44	1.84	2.69
Sample 43	3.07	2.62	3.21
Sample 44	2.13	2.67	2.95
Sample 45	2.79	2.76	2.88
Sample 46	2.95	2.23	3.48
Sample 47	2.66	1.44	2.50
Sample 48	2.46	2.29	2.53
Sample 49	2.74	2.54	3.29
Sample 50	2.30	2.25	2.41
Sample 51	2.51	2.68	3.51
Sample 52	2.55	2.36	1.91
Sample 53	2.91	2.89	3.15
Sample 54	2.17	2.53	2.87
Sample 55	3.34	2.30	2.81
Sample 56	2.44	2.24	3.14
Sample 57	3.14	3.18	3.08
Sample 58	2.69	2.28	3.56
Sample 59	2.25	2.16	2.23
Sample 60	2.24	2.52	2.95
Sample 61	2.16	2.36	2.05
Sample 62	2.59	2.89	3.61
Sample 63	2.66	2.06	3.39
Sample 64	2.36	2.60	3.48
Sample 65	2.44	3.32	3.06
Sample 66	3.39	2.71	2.85
Sample 67	2.84	1.82	4.15
Sample 68	2.63	2.05	2.70
Sample 69	3.04	2.52	2.99

TABLE 1-continued

Percentage of caffeine in dry matter from different resources in different seasons:			
Sample	Caffeine content (%)		
	Spring	Summer	Autumn
Sample 70	2.73	2.89	2.90
Sample 71	3.23	2.42	3.41
Sample 72	3.04	2.65	3.44
Sample 73	2.74	1.90	3.00
Sample 74	2.79	2.68	3.00
Sample 75	2.86	1.85	3.05
Sample 76	3.16	2.75	3.52
Sample 77	3.09	2.33	2.58
Sample 78	0.13	0.11	0.04
Sample 79	0.10	0.00	0.03
Sample 80	3.14	1.81	3.05
Sample 81	0.10	0.09	0.10
Sample 82	2.07	2.67	3.61
Sample 83	3.17	2.88	3.64
Sample 84	2.38	1.72	2.93
Sample 85	2.68	2.62	2.25
Sample 86	2.78	3.27	3.29
Sample 87	2.22	2.29	2.85
Sample 88	2.70	1.56	2.77
Sample 89	2.32	2.21	2.53
Sample 90	2.08	1.89	2.31
Sample 91	2.29	2.72	2.50
Sample 92	2.68	2.70	2.73
Sample 93	2.26	3.19	2.74
Sample 94	2.88	2.71	2.61
Sample 95	2.73	2.36	2.99
Sample 96	2.46	1.83	2.46
Sample 97	1.67	2.81	2.29
Sample 98	2.64	3.20	4.00
Sample 99	3.18	2.24	2.68
Sample 100	2.48	2.05	2.00
Sample 101	2.19	1.92	3.37
Sample 102	2.49	1.98	2.90
Sample 103	2.94	2.58	2.82
Sample 104	2.47	2.58	3.33
Sample 105	3.90	2.38	3.43
Sample 106	2.41	2.02	2.93
Sample 107	3.32	2.80	3.61
Sample 108	1.61	2.07	2.40
Sample 109	3.04	2.47	2.92
Sample 110	3.00	2.28	3.74
Sample 111	3.98	3.25	2.77
Sample 112	3.88	2.65	3.08
Sample 113	3.36	1.97	3.01
Sample 114	3.82	3.04	3.68
Sample 115	4.07	3.30	3.88
Sample 116	2.51	2.21	2.72
Sample 117	2.98	2.85	3.47
Sample 118	3.34	2.32	3.11
Sample 119	2.93	2.62	2.87
Sample 120	3.25	2.49	3.40
Sample 121	2.87	2.21	2.99
Sample 122	3.03	2.27	1.90
Sample 123	3.08	2.43	3.28
Sample 124	3.25	2.71	2.95
Sample 125	2.58	2.90	3.46
Sample 126	3.30	2.38	4.26
Sample 127	3.02	2.26	2.83
Sample 128	0.97	2.58	3.61
Sample 129	1.67	1.77	2.34
Sample 130	3.31	2.08	2.83
Sample 131	3.58	2.38	3.08
Sample 132	3.45	1.99	2.50
Sample 133	3.58	2.16	3.75
Sample 134	2.69	2.47	2.55
Sample 135	2.80	1.72	2.42
Sample 136	3.84	2.06	2.12
Sample 137	0.25	0.11	0.08
Sample 138	2.42	2.01	2.54
Sample 139	0.11	0.13	0.13
Sample 140	2.97	2.11	2.90
Sample 141	2.92	2.31	2.62
Sample 142	2.84	2.72	2.82

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TABLE 1-continued

Percentage of caffeine in dry matter from different resources in different seasons:			
Sample	Caffeine content (%)		
	Spring	Summer	Autumn
Sample 143	2.75	2.83	3.63
Sample 144	2.74	2.31	3.02
Sample 145	3.22	2.71	3.27
Sample 146	2.83	2.20	1.91
Sample 147	3.01	2.36	3.54
Sample 148	3.61	2.13	2.30
Sample 149	2.45	1.93	0.14
Sample 150	2.97	2.18	5.00
Sample 151	3.70	3.00	3.03
Sample 152	3.08	2.47	2.72
Sample 153	3.32	2.88	2.85
Sample 154	1.98	1.86	3.52
Sample 155	2.48	2.20	2.44
Sample 156	3.96	3.13	3.84
Sample 157	3.92	2.95	4.66
Sample 158	3.44	2.24	2.81
Sample 159	2.48	2.50	2.90
Sample 160	2.55	2.16	3.18
Sample 161	3.41	2.31	3.48
Sample 162	3.34	2.08	3.33
Sample 163	3.45	2.46	3.02
Sample 164	2.90	2.37	3.00
Sample 165	4.03	2.32	3.87
Sample 166	3.04	3.10	3.90
Sample 167	3.31	2.23	2.60
Sample 168	2.22	3.00	3.53
Sample 169	3.20	1.97	2.21
Sample 170	3.17	2.44	3.60
Sample 171	2.77	2.87	4.20
Sample 172	2.23	2.21	2.98
Sample 173	3.55	3.05	3.65
Sample 174	2.60	1.77	2.16
Sample 175	3.21	2.25	2.98
Sample 176	2.74	1.81	0.24
Sample 177	2.37	2.17	3.21
Sample 178	3.29	3.48	4.38
Sample 179	0.10	0.09	3.00
Sample 180	2.95	1.37	2.71
Sample 181	3.72	2.67	3.59
Sample 182	3.81	2.69	3.00
Sample 183	3.70	2.14	3.58
Sample 184	2.82	1.94	1.48
Sample 185	3.29	2.10	3.45
Sample 186	3.55	2.23	2.28
Sample 187	3.29	2.38	2.71
Sample 188	0.16	0.10	2.44
Sample 189	5.36	2.42	3.45
Sample 190	3.21	2.63	2.38
Sample 191	2.64	2.64	2.85

The variation of caffeine content in the population is shown in Table 2 and FIG. 1.

TABLE 2

Phenotypic variation in caffeine content:					
Season	Range (%)	Mean (%)	Standard deviation ^a SD	Coefficient of variation ^b CV	Diversity index ^c H'
Spring	0.09~5.36	2.72	0.81	0.30	1.90
Summer	0~3.48	2.29	0.64	0.28	1.85
Autumn	0.03~5.0	2.84	0.81	0.29	1.83

III. Association Analysis between Genotype and Traits

1. Experimental Procedure

The CTAB method was used to extract total DNA from buds of 191 tea plant resources, and it was ensured that A260/A280 of each DNA sample is between 1.8 and 2.0, and

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the concentration was greater than 100 µg/µl. The extracted DNA samples were used to detect genotypes located in the SNP site 1 (Scaffold4239:309117), the SNP site 2 (Scaffold115:803980), the SNP site 3 (Scaffold720:596655), the SNP site 4 (Scaffold3614:66549), the SNP site 5 (Scaffold349:3413816), and the SNP site 6 (Scaffold920:281727) of the "Shuchazao" CSS cultivar tea plant genome (<http://tpia.teaplant.org/index.html>), respectively. The association analysis of traits and markers was performed, significance level of the association was judged by p-value, and the p-value less than 1.25E-05 was the significance level.

2. Experimental Results

The p-values of the six SNP sites in different seasons are shown in Table 3.

TABLE 3

p-values of six SNP sites in different seasons			
Season	Spring	Summer	Autumn
Scaffold4239:309117	6.91E-10	7.33E-09	3.28E-05
Scaffold115:803980	6.07E-09	1.17E-13	9.38E-11
Scaffold720:596655	1.60E-15	4.41E-16	2.56E-13
Scaffold3614:66549	1.03E-11	3.15E-10	2.42E-08
Scaffold349:3413816	5.68E-09	3.51E-10	7.98E-09
Scaffold920:281727	6.12E-13	9.58E-12	4.13E-07

Embodiment 2 Verification of SNP Site

I. Experimental Method

Genotypes of the SNP site 1 (Scaffold4239:309117), the SNP site 2 (Scaffold115:803980), the SNP site 3 (Scaffold720:596655), the SNP site 4 (Scaffold3614:66549), the SNP site 5 (Scaffold349:3413816), and the SNP site 6 (Scaffold920:281727) were subjected to verification in another population of 98 germplasms.

1. Caffeine content of each sample was detected. The specific detection method is the same as that of Embodiment 1.
2. SnapShot technology platform was used to detect the genotypes of the SNP site 1 (Scaffold4239:309117), the SNP site 2 (Scaffold115:803980), the SNP site 3 (Scaffold720:596655), the SNP site 4 (Scaffold3614:66549), the SNP site 5 (Scaffold349:3413816), and the SNP site 6 (Scaffold920:281727) in each sample.

This method designed primers of different lengths for different mutation sites, after SNAPshot reaction, the products were analyzed by electrophoresis, five-color fluorescence detection, and Gene mapper analysis, and multiple SNP sites can be detected in one sequencing reaction.

SNAPshot was used for site-specific sequence analysis, and the basic principle thereof followed the dideoxy termination method in direct DNA sequencing, except that only ddNTPs with different fluorescent labels were used in the PCR reaction. Since the 3'-end of the primers of each SNP site is

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close to the SNP point, each of the primers was extended by only one nucleotide according to the sequence of the template under the action of the polymerase. Then an advanced fluorescence detection system was used to detect the type of that nucleotide that is extended.

(1) Design of primers

Primers were designed and synthesized according to the position of Scaffold4239:309117 in the genome. In particular, Scaffold4239:309117 each extended 500 bp upstream and downstream. A nucleotide sequence thereof is shown as SEQ ID NO: 1 (FIG. 2, wherein N denotes the base to be tested at Scaffold4239:309117).

PCR primers:

(SEQ ID NO: 2)
F: GAAGACTAACCCGTATCGAG;

(SEQ ID NO: 3)
R: ACACCTTACAGTCTCTTGCGG.

Single base extension primer:

(SEQ ID NO: 25)
ctgactgactgactgactgactATTGTCTCGTTGCTTCGGTTGTTTC.

Primers were designed and synthesized according to the position of Scaffold115:803980 in the genome. In particular, Scaffold115:803980 each extended 500 bp upstream and downstream. A nucleotide sequence thereof is shown as SEQ ID NO: 4 (wherein N denotes the base to be tested at Scaffold115:803980).

PCR primers:

(SEQ ID NO: 5)
F: CTTTCATCTCCACCACACTTC;

(SEQ ID NO: 6)
R: GCCCAAAGTAGCAAAGAGAG.

Single base extension primer:

(SEQ ID NO: 26)
gactgactgactgactgactgactcAGCAGAGCTTGGCAAAGAGGGATG.

Primers were designed and synthesized according to the position of Scaffold720:596655 in the genome. In particular, Scaffold720:596655 each extended 500 bp upstream and downstream. A nucleotide sequence thereof is shown as SEQ ID NO: 7 (FIG. 4, wherein N denotes the base to be tested at Scaffold720:596655).

PCR primers:

primer F: (SEQ ID NO: 8)
CAACTTTGGTGATGACGGAC;

primer R: (SEQ ID NO: 9)
TTCAACTGGTGTGTAGACGC.

Single base extension primer:

(SEQ ID NO: 27)
gactgactgactgactgactgactGCTACAGTTCGACTCGAATTGTCAC.

Primers were designed and synthesized according to the position of Scaffold3614:66549 in the genome. In particular, Scaffold3614:66549 each extended 500 bp upstream and

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downstream. A nucleotide sequence thereof is shown as SEQ ID NO: 10 (FIG. 5, wherein N denotes the base to be tested at Scaffold3614:66549).

PCR primers:

(SEQ ID NO: 11)
F: GATGACACAACCCCTCATCTG;

(SEQ ID NO: 12)
R: AATGTATGCCCGGTAAGGAC.

Single base extension primer:

(SEQ ID NO: 28)
gactACTAACTTTACGCCACGACCCA.

Primers were designed and synthesized according to the position of Scaffold349:3413816 in the genome. In particular, Scaffold349:3413816 each extended 500 bp upstream and downstream. A nucleotide sequence thereof is shown as SEQ ID NO: 13 (FIG. 6, wherein N denotes the base to be tested at Scaffold349:3413816).

PCR primers:

primer F: (SEQ ID NO: 14)
TCTCTGCACTGTTGTCACTC;

primer R: (SEQ ID NO: 15)
CACCACACTTTCTTAGAAGG.

Single base extension primer:

(SEQ ID NO: 29)
actgactgactaAGGATCTAGTCCCTGCATAAATAACA.

Primers were designed and synthesized according to the position of Scaffold920:281727 in the genome. In particular, Scaffold920:281727 each extended 500 bp upstream and downstream. A nucleotide sequence thereof is shown as SEQ ID NO: 16 (FIG. 7, wherein N denotes the base to be tested at Scaffold920:281727).

PCR primers:

primer F: (SEQ ID NO: 17)
TTGCATTCTGCTCTTTTGGG;

primer R: (SEQ ID NO: 18)
ACGTGCTACATTCTCCATCC.

Single base extension primer:

(SEQ ID NO: 30)
tgactgactgactgactgactgactgactTAGCATCTAAGAAAGAG
GATTTA.

(2) PCR Amplification

PCR system (10 μ l) was as follows:

2 $\times$ Taq PCR Master Mix	5 μ l
PrimerMix (matching according to the amplification ratio)	1 μ l

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-continued

DNA template	1 µl
ddH ₂ O	3 µl

PCR amplification procedure was as follows:

95° C.	5 minutes	×45 cycles
95° C.	30 seconds	
56° C.	30 seconds	
72° C.	30 seconds	
72° C.	2 minutes	
4° C.	forever	

(3) PCR Product Purification

Purification was performed using shrimp alkaline phosphatase purification. The main functional components of shrimp alkaline phosphatase MIX (EX-SAP) are SAP and ExoI. SAP enzyme, which can dephosphorylate residual dNTPs, and ExoI degrades the free single-chain primer. 4 µl of PCR product was taken and added with 2 µl of EX-SAP enzyme. The specific reaction system is shown as follows:

Constituent of digestive system	Volume (µl)
ddH ₂ O	0.75
SAP (1 U/µl)	0.5
ExoI (5 U/µl)	0.15
10*SAP buffer	0.6
PCR product	4
Total volume	6

After that, digestion and incubation were performed on a PCR instrument: 37° C. for 40 minutes, 85° C. for 5 minutes, 4° C. forever.

(4) SNAPshot Reaction

The PCR product was used as a template for SNAPshot reaction.

The SNAPshot reaction system is shown as follows:

Reagent	Dosage (µl)
SNaPshot Mix	0.5
Pooled PCR Products	3

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-continued

Reagent	Dosage (µl)
Pooled Primers	1
dH ₂ O	0.5
Total volume	5

The SNAPshot reaction procedure is:

95° C.	2 minutes	×40 cycles
95° C.	10 seconds	
52° C.	5 seconds	
60° C.	30 seconds	
4° C.	forever	

After that, the SNAPshot product was purified, and 2 µl of SAP mix was directly added to the SNAPshot reaction product. The specific reaction system was as follows:

Constituent	Volume (µl)
Water	0.9
SAP(1 U/µl)	0.5
10*SAP buffer	0.6
Total	2

The SNAPshot product digestion reaction was performed on a PCR instrument, and the reaction procedures were: 37° C. for 40 minutes, 75° C. for 15 minutes, 4° C. forever.

(5) On-Machine Detection

2 µl of the digested SNAPshot reaction product was taken and added into 8 µl of deionized formamide containing 0.4% LIZ120, denatured at 95° C. for 5 minutes, then quenched at -20° C., and then sequenced on 3730XL.

(6) Result analysis

The .fsa results obtained by GeneMarker analysis were used to derive peak plots and table files, and to calculate the SNP mutant type of each sample

II. Experimental Results

Caffeine content and genotypes of SNP1, SNP2, SNP3, SNP4, SNP5, SNP6 sites of each sample are shown in Table 4, and the SNaPshot sequencing results of some samples are shown in FIG. 8 to FIG. 24.

TABLE 4

The CAF content in dry matter and genotype of the resource in the population:

Sample	Caffeine content (%)	SNP1 genotype	SNP2 genotype	SNP3 genotype	SNP4 genotype	SNP5 genotype	SNP6 genotype
Sample 2-1	2.36	GA	AA	CC	CT	AA	AA
Sample 2-2	2.89	GA	AA	CC	CC	AA	AA
Sample 2-3	2.63	GG	AA	CC	TT	AA	AA
Sample 2-4	3.38	GG	AA	CC	TT	AA	AA
Sample 2-5	2.82	GG	AA	CC	CT	AA	AA
Sample 2-6	2.24	GG	AA	CC	TT	AA	AA
Sample 2-7	2.36	GG	AA	CC	TT	AA	AA
Sample 2-8	2.18	GG	AA	CC	TT	AA	AA
Sample 2-9	2.06	GG	AA	CC	TT	AA	AA
Sample 2-10	2.93	GG	AA	CC	TT	AA	AA
Sample 2-11	2.58	AA	AA	CC	CT	AA	AA
Sample 2-12	2.52	GG	AA	CC	TT	GA	AA
Sample 2-13	2.67	GA	AA	CC	CT	AA	AA
Sample 2-14	0.12	AA	AA	CC	CC	AA	AA
Sample 2-15	3.17	GG	AA	CC	TT	AA	AA

TABLE 4-continued

The CAF content in dry matter and genotype of the resource in the population:							
Sample	Caffeine content (%)	SNP1 genotype	SNP2 genotype	SNP3 genotype	SNP4 genotype	SNP5 genotype	SNP6 genotype
Sample 2-16	3.05	GG	AA	CC	TT	AA	AA
Sample 2-17	2.51	GG	AA	CC	TT	AA	AA
Sample 2-18	1.76	AA	AA	CC	CT	AA	AA
Sample 2-19	3.06	GG	AA	CC	TT	GA	AA
Sample 2-20	2.42	AA	AA	CC	CT	AA	AA
Sample 2-21	2.73	GG	AA	CC	TT	AA	AA
Sample 2-22	2.84	GG	AA	CC	TT	AA	Not detected
Sample 2-23	2.54	GG	AA	CC	TT	AA	AA
Sample 2-24	2.61	GA	AA	CC	CT	AA	AA
Sample 2-25	2.82	GG	AA	CC	TT	AA	AA
Sample 2-26	2.28	GA	AA	CC	CT	AA	AA
Sample 2-27	2.20	GA	GA	CC	TT	AA	AA
Sample 2-28	2.21	GG	AA	CC	TT	AA	Not detected
Sample 2-29	2.16	GG	AA	CC	TT	AA	AA
Sample 2-30	2.43	GG	AA	CC	TT	AA	AA
Sample 2-31	1.67	GG	AA	CC	TT	AA	AA
Sample 2-32	2.10	GG	AA	CC	TT	GA	Not detected
Sample 2-33	1.95	GG	AA	CC	TT	AA	AA
Sample 2-34	2.46	GG	AA	CC	TT	AA	AA
Sample 2-35	2.80	GG	AA	CC	TT	AA	AA
Sample 2-36	2.31	GG	AA	CC	TT	GA	AA
Sample 2-37	2.79	GA	AA	CC	TT	AA	AA
Sample 2-38	2.19	GG	AA	CC	CC	GA	Not detected
Sample 2-39	2.08	GA	AA	CC	TT	AA	AA
Sample 2-40	2.72	GA	AA	CC	TT	AA	AA
Sample 2-41	3.10	GG	AA	CC	TT	AA	AA
Sample 2-42	2.99	GG	AA	CC	TT	AA	AA
Sample 2-43	2.12	GA	AA	CC	TT	AA	AA
Sample 2-44	2.94	AA	AA	CC	CT	AA	AA
Sample 2-45	2.76	GG	AA	CC	TT	GA	Not detected
Sample 2-46	3.01	GA	GA	CC	TT	AA	AA
Sample 2-47	2.21	GG	AA	CC	TT	AA	AA
Sample 2-48	2.75	GG	AA	CC	TT	AA	Not detected
Sample 2-49	2.12	GG	AA	CC	TT	GA	AA
Sample 2-50	2.67	GG	GA	CC	TT	AA	AA
Sample 2-51	2.24	GA	AA	CC	TT	AA	AA
Sample 2-52	2.56	GG	AA	CC	TT	AA	AA
Sample 2-53	3.29	GG	AA	CC	TT	AA	AA
Sample 2-54	3.06	GG	AA	CC	TT	AA	AA
Sample 2-55	2.43	GG	AA	CC	TT	AA	AA
Sample 2-56	2.63	GG	AA	CC	TT	AA	AA
Sample 2-57	2.74	GG	AA	CC	TT	AA	AA
Sample 2-58	2.74	GG	AA	CC	TT	AA	AA
Sample 2-59	3.24	GG	AA	CC	TT	AA	AA
Sample 2-60	3.29	GG	AA	CC	TT	AA	AA
Sample 2-61	2.63	GG	AA	CC	TT	AA	AA
Sample 2-62	2.92	GG	AA	CC	TT	AA	AA
Sample 2-63	1.65	AA	AA	CC	TT	AA	AA
Sample 2-64	1.44	GG	AA	CC	TT	AA	AA
Sample 2-65	1.94	GG	AA	CT	TT	AA	Not detected
Sample 2-66	2.38	GG	AA	CC	TT	GG	AA
Sample 2-67	2.52	GG	AA	CC	TT	AA	AA
Sample 2-68	2.42	GG	AA	CC	TT	GA	AA
Sample 2-69	2.48	GG	AA	CC	CT	AA	AA
Sample 2-70	2.61	GG	AA	CC	TT	AA	Not detected
Sample 2-71	2.94	GG	AA	CC	TT	AA	AA
Sample 2-72	2.12	GA	AA	CC	CT	AA	AA
Sample 2-73	2.66	GA	AA	CC	CT	AA	AA
Sample 2-74	2.45	GG	AA	CC	TT	AA	Not detected
Sample 2-75	2.41	AA	AA	CC	CT	AA	AA
Sample 2-76	2.57	GG	AA	CC	TT	AA	AA
Sample 2-77	2.64	GA	GA	CT	CC	GG	GG
Sample 2-78	2.17	GG	AA	CC	TT	AA	AA
Sample 2-79	1.88	GG	AA	CC	TT	AA	AA
Sample 2-80	2.01	GA	GA	CC	TT	AA	AA
Sample 2-81	2.44	GG	AA	CC	TT	AA	AA
Sample 2-82	2.26	GG	AA	CC	TT	AA	AA
Sample 2-83	2.39	GA	AA	CC	TT	AA	AA
Sample 2-84	2.39	GG	AA	CC	TT	AA	AA
Sample 2-85	2.95	GG	AA	CC	CT	AA	AA
Sample 2-86	2.86	GA	AA	CC	TT	AA	AA
Sample 2-87	2.87	GA	AA	CC	TT	AA	AA
Sample 2-88	3.04	GA	AA	CC	TT	AA	AA
Sample 2-89	3.28	GA	AA	CC	TT	AA	AA

TABLE 4-continued

The CAF content in dry matter and genotype of the resource in the population:							
Sample	Caffeine content (%)	SNP1 genotype	SNP2 genotype	SNP3 genotype	SNP4 genotype	SNP5 genotype	SNP6 genotype
Sample 2-90	3.16	GG	AA	CC	TT	AA	AA
Sample 2-91	2.09	GG	AA	CC	TT	AA	AA
Sample 2-92	2.85	GG	AA	CC	TT	AA	Not detected
Sample 2-93	2.63	GG	AA	CC	TT	AA	AA
Sample 2-94	1.02	GG	AA	CC	CT	AA	GA
Sample 2-95	3.21	GG	AA	CC	CT	GA	AA
Sample 2-96	2.30	GG	AA	CC	TT	AA	AA
Sample 2-97	0.07	AA	GG	CT	CC	GG	GG
Sample 2-98	0.08	AA	GA	CT	CC	GG	GG

The significance analysis results show that the genotype of Scaffold4239:309117 is extremely significantly correlated with caffeine content, the correlation coefficient is -0.4 , p-value is 3.66×10^{-5} , F-value (6.91/3.94) is 18.7, which is a recessive mutation, and the caffeine content in the dry matter of tea soup corresponding to an AA genotype sample has extremely significant difference compared with GG and GA genotype samples. It is statistically judged that, when the genotype sample is double mutant AA, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type GG or single mutant GA.

The significance analysis results show that, the genotype of Scaffold115:803980 is extremely significantly correlated with caffeine content, the correlation coefficient is -0.4 , p-value is 3.66×10^{-5} , F-value (6.91/3.94) is 18.7, which is a recessive mutation, and the caffeine content in the dry matter of the tea plant corresponding to a GG genotype sample has extremely significant difference compared with AA and GA genotype samples. It is statistically judged that, when the genotype of the sample is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

The significance analysis results show that, the genotype of Scaffold720:596655 is extremely significantly correlated with caffeine content, the correlation coefficient is -0.51 , p-value is 5.78×10^{-6} , F-value (6.91/3.94) is 23.1, which is a dominant mutation, the caffeine content in the dry matter of the tea plant corresponding to a wild type CC genotype sample has significant difference compared with a single mutant CT genotype sample. It is statistically judged that, when the genotype of the sample is single mutant CT genotype, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample with wild type CC genotype.

The significance analysis results show that, the genotype of Scaffold3614:66549 is extremely significantly correlated with caffeine content, the correlation coefficient is -0.48 , p-value is 5.45×10^{-7} , F-value (6.91/3.94) is 28.9, which is a recessive mutation, and the caffeine content in the dry matter of tea soup corresponding to a CC genotype sample has significant difference compared with TT and CT genotype samples. It is statistically judged that, when the genotype is double mutant CC, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type TT or single mutant CT.

The significance analysis results show that, the genotype of Scaffold349:3413816 is extremely significantly correlated

lated with caffeine content, the correlation coefficient is -0.4 , p-value is -4.04×10^{-5} , F-value (6.91/3.94) is 18.5, which is a recessive mutation, the caffeine content in the dry matter of tea soup corresponding to a GG genotype sample has significant difference compared with GA and AA genotype samples. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

The significance analysis results show that, the genotype of Scaffold920:281727 is extremely significantly correlated with caffeine content, the correlation coefficient is -0.45 , p-value is 3.16×10^{-6} , F-value (6.91/3.94) is 18.7, which is a recessive mutation, the caffeine content in the dry matter of tea soup corresponding to a GG genotype sample has significant difference compared with GA and AA genotype samples. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

Embodiment 3 Kit for Evaluating Tea Plant Caffeine Content

I. Composition

The primers for the SNP site 1 which have the nucleotide sequence shown as SEQ ID NO: 2 and SEQ ID NO: 3, the primers for the SNP site 2 which have the nucleotide sequence shown as SEQ ID NO: 5 and SEQ ID NO: 6, the primers for the SNP site 3 which have the nucleotide sequence shown as SEQ ID NO: 8 and SEQ ID NO: 9, the primers for the SNP site 4 which have the nucleotide sequence shown as SEQ ID NO: 11 and SEQ ID NO: 12, the primers for the SNP site 5 which have the nucleotide sequence shown as SEQ ID NO: 14 and SEQ ID NO: 15, the primers for the SNP site 6 which have the nucleotide sequence shown as SEQ ID NO: 17 and SEQ ID NO: 18, 2×Taq PCR Master Mix, ddH₂O.

In particular, primer F for SNP site 1:
(SEQ ID NO: 2)

GAAGACTAACCCGTATCGAG;

primer R for SNP site 1:
(SEQ ID NO: 3)

ACACTTACAGTCTCTTGCGG;

primer F for SNP site 2:

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-continued

CTTCATCTCCACCACACTTC; (SEQ ID NO: 5)

primer R for SNP site 2: (SEQ ID NO: 6) 5

GCCCAAAGTAGCAAGAGAG; (SEQ ID NO: 8)

primer F for SNP site 3: (SEQ ID NO: 9)

CAACTTTGGTGATGACGGAC; (SEQ ID NO: 11)

primer R for SNP site 3: (SEQ ID NO: 12)

TTCAACTGGTGATGACGC; (SEQ ID NO: 14)

primer F for SNP site 4: (SEQ ID NO: 15)

GATGACACAACCCTCATCTG; (SEQ ID NO: 17)

primer R for SNP site 4: (SEQ ID NO: 18)

AATGTATGCCCGTAAGGAC; (SEQ ID NO: 19)

primer F for SNP site 5: (SEQ ID NO: 20)

TCTCTGCACTGTTGCTACTC; (SEQ ID NO: 21)

primer R for SNP site 5: (SEQ ID NO: 22)

CACCACACTTTCTTAGAAGG; (SEQ ID NO: 23)

primer F for SNP site 6: (SEQ ID NO: 24)

TTCGCATTCTGCTTTTGGG; (SEQ ID NO: 25)

primer R for SNP site 6: (SEQ ID NO: 26)

ACGTGCTACATTCTCCATCC.

II. Usage method

(1) The CTAB method was used to extract total DNA from buds of tea plant, and it was ensured that A260/A280 of each DNA sample is between 1.8 and 2.0, and the concentration was greater than 100 µg/µl.

(2) PCR Amplification

Detection primers with nucleotide sequences shown as SEQ ID NO: 2 and SEQ ID NO: 3, SEQ ID NO: 5 and SEQ ID NO: 6, SEQ ID NO: 8 and SEQ ID NO: 9, SEQ ID NO: 11 and SEQ ID NO: 12, SEQ ID NO: 14 and SEQ ID NO: 15 and SEQ ID NO: 17 and SEQ ID NO: 18 were used for detecting SNP site 1, SNP site 2, SNP site 3, SNP site 4, SNP site 5 and SNP site 6, respectively.

PCR system (10 µl) was as follows:

2x Taq PCR Master Mix	5 µl
primer	Each 0.5 µl
DNA template	1 µl
ddH ₂ O	3 µl

PCR amplification procedure was as follows:

95° C.	5 minutes	x45 cycles
95° C.	30 seconds	
56° C.	30 seconds	
72° C.	30 seconds	
72° C.	2 minutes	
4° C.	forever	

(3) Product Purification

The PCR amplification products were subjected to gel electrophoresis, followed by recovery and purification using a commercially available gel electrophoresis DNA recovery kit.

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A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 2 and SEQ ID NO: 3 was selected for recovery and purification.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 5 and SEQ ID NO: 6 was selected for recovery and purification.

A band with a fragment length of about 250 bp in the amplification product of the primers shown in SEQ ID NO: 8 and SEQ ID NO: 9 was selected for recovery and purification.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 11 and SEQ ID NO: 12 was selected for recovery and purification.

A band with a fragment length of about 240 bp in the amplification product of the primers shown in SEQ ID NO: 14 and SEQ ID NO: 15 was selected for recovery and purification.

(4) Sequencing and Interpretation of Results

The amplification products of the primers shown in SEQ ID NO: 2 and SEQ ID NO: 3 were recovered and purified and sent to a sequencing company for Sanger sequencing. The sequencing results were compared with the nucleotide sequence shown in SEQ ID NO: 1. According to FIG. 2 (bold and underlined parts denote upstream and downstream primers), the site Scaffold4239:309117 is located at the 73rd base of the amplification product. It is statistically judged that, when the genotype sample is double mutant AA, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type GG or single mutant GA.

The amplification products of the primers shown in SEQ ID NO: 5 and SEQ ID NO: 6 were recovered and purified and sent to a sequencing company for Sanger sequencing. The sequencing results were compared with the nucleotide sequence shown in SEQ ID NO: 4. According to FIG. 3 (bold and underlined parts denote upstream and downstream primers), the site Scaffold115:803980 is located at the 164th base of the amplification product. It is statistically judged that, when the genotype of the sample is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

The amplification products of the primers shown in SEQ ID NO: 8 and SEQ ID NO: 9 were recovered and purified and sent to a sequencing company for Sanger sequencing. The sequencing results were compared with the nucleotide sequence shown in SEQ ID NO: 7. According to FIG. 4 (bold and underlined parts denote upstream and downstream primers), the site Scaffold720:596655 is located at the 189th base of the amplification product. It is statistically judged that, when the genotype of the sample is single mutant CT genotype, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample with wild type CC genotype.

The amplification products of the primers shown in SEQ ID NO: 11 and SEQ ID NO: 12 were recovered and purified and sent to a sequencing company for Sanger sequencing. The sequencing results were compared with the nucleotide sequence shown in SEQ ID NO: 10. According to FIG. 5 (bold and underlined parts denote upstream and downstream primers), the site Scaffold3614:66549 is located at the 137th base of the amplification product. It is statistically judged that, when the genotype is double mutant CC, the caffeine content in the dry matter in the tea plant is more likely to be

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lower than the normal average of the sample of which the genotype is wild type TT or single mutant CT.

The amplification products of the primers shown in SEQ ID NO: 14 and SEQ ID NO: 15 were recovered and purified and sent to a sequencing company for Sanger sequencing. The sequencing results were compared with the nucleotide sequence shown in SEQ ID NO: 13. According to FIG. 6 (bold and underlined parts denote upstream and downstream primers), the site Scaffold349:3413816 is located at the 160th base of the amplification product. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA.

The amplification products of the primers shown in SEQ ID NO: 17 and SEQ ID NO: 18 were recovered and purified and sent to a sequencing company for Sanger sequencing. The sequencing results were compared with the nucleotide sequence shown in SEQ ID NO: 16. According to FIG. 7

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(bold and underlined parts denote upstream and downstream primers), the site Scaffold920:281727 is located at 106th base of the amplification product. It is statistically judged that, when the genotype is double mutant GG, the caffeine content in the dry matter in the tea plant is more likely to be lower than the normal average of the sample of which the genotype is wild type AA or single mutant GA

Embodiment 4 Use of Kit for Evaluating Tea Plant Caffeine Content

I. Experimental Method

The kit in Embodiment 3 was used to detect 98 tea plant samples in Embodiment 2.

II. Experiment Results

The detection results are consistent with those of Embodiment 2 using the SnapShot technology platform. This kit can be used to evaluate the tea plant caffeine content. The sequencing peaks of some samples are shown in FIG. 25 to FIG. 29.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 30

<210> SEQ ID NO 1

<211> LENGTH: 1001

<212> TYPE: DNA

<213> ORGANISM: Camellia sinensis

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (501)..(501)

<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 1

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gaaggctctg gagtagctga agttgttatg agcttgtcta ggccgaaatc agcgaggtga      60
gcttcaaaat cggcgctcgaa taggacgttc tgaggcttga catcgccatg aaccatggcg     120
gtggagtgga ggaaggcgag gcccgggcg attccgaggg ctattaggtg gcgcattggc     180
caattcaata catgccctgc ttggtgagaa gcttcttgaa gcaatgtggc taggtttccg     240
ttaggcatat agtcgtagac taagagtctg aggtctgggt gtccggcgaa gtaccacgg      300
aggactgtga ggtttctgtg cttcactctc ccgagcgatt cggcttcttt tctgaacatg     360
ttttcgtcta gcgatccatc agggagtctc cgaatcgaaa gcaccattcc atcactgtaa     420
caggctttga agactaacco gtatcgagtc ctgcttagaa cgttctcttc atcgaattgt     480
ctcgttgctt cggttgtttc ngctagagtg atcttggtat tgaacataac aagcttttga     540
ccgccattat ccgcaacttc acgacctccg ctggtgcgag ctgagcttgc tcttctggg      600
ctgcgctttt tctctccggc agccttttct ttgagcctct tgcgccaccg caagagactg     660
taagtgtaga agcaacaaca cagtgcctaag aggaaccac cactaacagc catggcaata     720
aacatgatca gcctcttctt cctattactc atctcttcgc atttcgtgct taagggtttc     780
ccacataagt tcggatttcc tgcataatca gatggatcgt tgaatcttga agccagcatt     840
gttggaatct cgcgggagag gttgttttgg gatacattga agtagaccaa gctagagatg     900
agtgaatgtt ttgctggaat cggtcgggtc aggttgtttg cagagagatt gaggactgtg     960
aggtttgata aattggacaa tgagtctggt atttggcctg g                          1001

```

<210> SEQ ID NO 2

<211> LENGTH: 20

<212> TYPE: DNA

<213> ORGANISM: Camellia sinensis

-continued

<400> SEQUENCE: 2

gaagactaac ccgtatcgag

20

<210> SEQ ID NO 3

<211> LENGTH: 20

<212> TYPE: DNA

<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 3

acacttacag tctcttgagg

20

<210> SEQ ID NO 4

<211> LENGTH: 1001

<212> TYPE: DNA

<213> ORGANISM: Camellia sinensis

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (501)..(501)

<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 4

aatcattaag agtcattatg gtaatcatga gcttaattac tccaagtaaa gccaatcttc 60

atcatagaaa taaaatttac aaaaaaaaaa aaaaaaaaaa agtctttcag ctgaacaacc 120

catccctgca actgcaccac cataattgag atctaaatct gaaggaactt gcttgagatc 180

taaattctgaa ggaacttgct tgcttaggaa catccacatc catgatttct acaatttttg 240

gaagacacag aaccagagaa gatgactcaa aatcaagcag caattgtaag aaaattcgac 300

caatcgaaat catcttgga ttaatcattg tagcctcctt catctccacc acacttctcc 360

tcctacttcc atgcgattac gtcgacggca gccctattcc caccatcata ttcaaaggac 420

tccccccac cttccacgcc ttgcgtcgtc cctcatctt cgccttctcc ggagccttga 480

ggcgttggtt gatccacgac ncatccctct ttgccaagct ctgcgagttc tcttccatgg 540

cctccatgac ctctgctctc tctttgctac tttgggctat gttcttcacc tgttttcaac 600

cacaaccag gtaaaactcg aattcagaca tcacatggta agaaaacaag ttattaagggt 660

ttttaacctt ataaagactt tttttctttt tttttttcct tctgttccaa cggacacgtg 720

gtgtgtttta aaattaataa atcgtgtatc agatatggat atacaatcgc gtggtcagtt 780

gaaattacta ttggtatgct ttatataccg tgcgtgtgtt aaaattaaaa cttgttttgt 840

gatgttggtg gtctgttatg tacttggtgt tgttgaaata atattaccat aaatttgaat 900

aagcctttat tatgtggaga tccgatggat taatgatgca tattgtcaca gaattcaaaa 960

tgatttcatt ttgagcatgg tgacgagggt tccaagccct g 1001

<210> SEQ ID NO 5

<211> LENGTH: 20

<212> TYPE: DNA

<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 5

cttcatctcc accacacttc

20

<210> SEQ ID NO 6

<211> LENGTH: 20

<212> TYPE: DNA

<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 6

-continued

gcccaaagta gcaaagagag 20

<210> SEQ ID NO 7
 <211> LENGTH: 1001
 <212> TYPE: DNA
 <213> ORGANISM: Camellia sinensis
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (501)..(501)
 <223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 7

```

gccactaga taatggcaaa accatctttt ctaaagccca cttgagggtg gcaaaatctg    60
cattttagcc tacttgcgag tagcaaacca ttttcagaac aaacatgaaa tgttttgttc    120
caatttaggc gatttggctt acacttggtt gctgtgcgca taacatagta attctaggct    180
aataaccgga ataaaagcat ataattgttt caaaacaaga tatcatggtt agtagaggcc    240
gatctttgat cgggagagtg ggtatgctca tagctgacct gcccgaattg tacccaagtt    300
cccgagccga gacaactttg gtgatgacgg acacgtgccc tttggcaccg agtcagggtt    360
tcttaaatto tcccaaaata aaatathtag gtggcgactc tgtatctggc aaagcagtcc    420
atgtttggca cttcttttct aaaaatctgt ttttcacaaa caatttgccg gatttggact    480
ctttggttca gattaagaac ngtgacaatt cgagtcgaa ctgtagcatg gggccacagg    540
gcgcgtctac acaccagttg aaggtaaaga cagaagatgt tgagaaaatt gcattcagag    600
cgaggtatgg tcattatgaa ttccttggtt tgccatttgg agtaactaat gcccctgcaa    660
cgtttataga cctaataaac tgtattttta agacttatct tgatgatttt gttgtgattt    720
tattaatgat atcttggtgt attcaaagaa tagacttgaa catgaacacc acttgagaac    780
tttcttgcaa acacttaaag aaaagaaatt attttccaaa ctaaaaaaat gtgaattttg    840
gttggatgaa gtcattttct tagggcatgt tatcaacaag gaagaaatth cagtagatcc    900
acagaaaatt gaagcaattg tgaattgacc cactccaaca aatataacag aagtccatag    960
tttcgtgggc ttactgggtt actacagaag atttgtgaag g                                1001

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<210> SEQ ID NO 8
 <211> LENGTH: 20
 <212> TYPE: DNA
 <213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 8

```

caactttggt gatgacggac    20

```

<210> SEQ ID NO 9
 <211> LENGTH: 20
 <212> TYPE: DNA
 <213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 9

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ttcaactggt gtgtagacgc    20

```

<210> SEQ ID NO 10
 <211> LENGTH: 1001
 <212> TYPE: DNA
 <213> ORGANISM: Camellia sinensis
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (501)..(501)
 <223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 10

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gagtcacggg tttcttaaat ttctctaaaa aatatttagg tggcgactct gtatctggca    60
aaatagtcga tttttggcaa ttgattcaa aatcagtttt ccaacatatt tgccgaattg    120
ggactttttg gtgattatct atttcacatt gcacatgtga aatcagattc agaaccgtgg    180
gagtcacgata ctgtagggct tattcgtctt ccgaaaaggg gcacgcaaag tcgaactaca    240
agtccctcgg ggaggatgga ttgcaaaatt accgtacaca gtagcaatcc cgtctttaa    300
ggcgactttt accaactgat ggaccattga tgacacaacc ctcatctgat gtagccaggg    360
tcttccagct agtagattga aagtgtccga aacatccatg acatagaatt taacctgatg    420
ctcagacggg ccgagtagga tatggctctt aaacattacc atgacatctt ggctcgtatt    480
gtcatataag cctaaacggc ntgggtcgtg ggcgtaaagt tagtcggcct cacaccgatg    540
gcacagggcg tccttacggg gcatacatta atcgccgacg cgttatctac caacaccact    600
ggaatccact ttttttgact ttccagcgtt acatataagg gccaatgtg gttagcacc    660
tcagggtgta actctttatc tataaaagat atcactggcg taacatcccc ggatgtaacc    720
aatgatacca attggtcagc agtgggttcg atagggagtt tggtcgggtt cattgcctct    780
agcagcagtg cctgtctatg ctcccgagat gccatgatta gccccagat tgatatgtcg    840
gcctgaatct tottaagctg tttcaagacc aggttttctt caacatcctt ctcttttgat    900
ttctcgaccc ccaactgtct tgatatatgc catcttttag gggtatcacc cattggtacc    960
cctttcggtc tagattaccc tgactttaag gtctccttct c                                1001

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<210> SEQ ID NO 11
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

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<400> SEQUENCE: 11

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gatgacacaa ccctcatctg                                20

```

```

<210> SEQ ID NO 12
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

```

```

<400> SEQUENCE: 12

```

```

aatgtatgcc cggtaaggac                                20

```

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<210> SEQ ID NO 13
<211> LENGTH: 1001
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (501)..(501)
<223> OTHER INFORMATION: n is a, c, g, or t

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<400> SEQUENCE: 13

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ataatctttt tgtacttggt cagggtggaat gaagcaatca accgagagtc caggaacatt    60
gaatgctagg tcgtogatct tccaagtctc ctccatccgt gtgattgctg tgcccgtct    120
cagattgtcc ccaaatcttg agatgatcac acttgattgg ccagaatgcg cgatcatcgt    180
gcccctcacc atccgatagt cctcgatttt cgtgcccacg gtggtctccc aataggtagg    240
gtaggttccg ggggactgga ttctggtgag gtaagagtc tctaaataca ctacgagacc    300
acttcttttg ctgaagtaac caaacatgac atgcttgatc atctctgcac tgttgtoact    360

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-continued

ccgatcggct aggtccgtct gatccgcgga caatttcaac acgaagcaat cgacgctcaa	420
gattcgtttt tcgcccacgt attgtgctag ggaaacaca gccgatacag ccacaggatc	480
tagtcctgc ataaataaca ntatgtttt tacatagagg aaaataatat ctgtcacatg	540
aattctactc ctttttttaa ctttctaaga aagtgtggtg aaaaaaatat taaatccatt	600
gggtaaaata taacagtcct taacataaca atatggcgaa ctatacattc aattctagaa	660
aatgtctcat ttttatagat ttttatgaaa gggatcaacc ttctttttt ttattggaag	720
cactatataa ataatgtcaa atagttttcc aaacttatct aaataaagtt ttaataattt	780
taatccacac attttgaatt taatttactt attttttagt gataacatta ccacagtcaa	840
aaagagtgcc aacatgaacc tccagcacac ttgaagagca cttgacgac atattgggaa	900
agttaccagc cagcactccc aaaaaaaaaa aaagaaaaaa agataaaaga ttaaaaaaat	960
tagtaaaaag tgactttaca aaaaggaata ttccacctct g	1001

<210> SEQ ID NO 14
 <211> LENGTH: 20
 <212> TYPE: DNA
 <213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 14

tctctgcact gttgtcactc	20
-----------------------	----

<210> SEQ ID NO 15
 <211> LENGTH: 20
 <212> TYPE: DNA
 <213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 15

caccacactt tcttagaagg	20
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<210> SEQ ID NO 16
 <211> LENGTH: 1001
 <212> TYPE: DNA
 <213> ORGANISM: Camellia sinensis
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (501)..(501)
 <223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 16

agggagactt ttatcttgag agctagaaga agagaaagtt agagaaaaga aagagaagta	60
ggaagaaaat caaagggaat tcacattcgt ccttttgagg ttgagaattg aacacttagg	120
tgatttcgaa aatcataaat gaggtgtgtt aaactaatat cgttcagcta cagttactca	180
gtaaattctc tttctcagag gctacgcagg tgtagtttga gttaaacttg gccacttaaa	240
ctaattggaac cattaggggc ccaagctaatt tagttcctag aacaaaggag agaggacgga	300
gaagcataga gaaagttaga gagaaacttt tttcttgaga gatagaagag atagttagag	360
aaaagaaaga gaaacgggaa aaaaatcatt gggaattcgc attcgtcctt ttgggcttga	420
gaattgaaca gttggggaat ttgggaaacc ttaaatgcgg tgcttatgtt taactaatat	480
cgttaagtgc caattactca ntaaatcctc tttcttagat gctaagcaag atttagtgta	540
gttaaacttg gccacttaag ctaattggaac agttagggtc ccaagcgaat tagtttccta	600
gaacaaaaga tagaaggatg gagaatgtag caggttcgtg agggaccccg ctactacagt	660
tcggactcga tttgtgtcac ggttcttaat ctgaacaaa gagtccaaat ccggcaaatc	720
gttttgagaa acagattttt tgaaaagaag tgccaaacat ggactgcttt gctagatata	780

-continued

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gagtcgccac ctaaatat ttttaaaatg gggaaattta ggaaacccta acttggtgcc      840
aaaggccacg tgtccgtcat tgccaaagt gcttgggctc gggagcttgg gtacgattgg      900
ggaaggctcag ctatgagcac cccctctcgc ccgatccgaa gatcggcctc tactaaccgt      960
gatatccggtt ttgaaaaacg ttatgtgttc ttaaaccaat t                          1001

```

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<210> SEQ ID NO 17
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

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<400> SEQUENCE: 17

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ttcgcatctg tccttttggg      20

```

```

<210> SEQ ID NO 18
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

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<400> SEQUENCE: 18

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acgtgctaca ttctccatcc      20

```

```

<210> SEQ ID NO 19
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

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```

<400> SEQUENCE: 19

```

```

ccgcaagaga ctgtaagtgt      20

```

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<210> SEQ ID NO 20
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

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<400> SEQUENCE: 20

```

```

ctctctttgc tactttgggc      20

```

```

<210> SEQ ID NO 21
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

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```

<400> SEQUENCE: 21

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```

gcgtctacac accagttgaa      20

```

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<210> SEQ ID NO 22
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

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<400> SEQUENCE: 22

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gtccttacgg ggcatacatt      20

```

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<210> SEQ ID NO 23
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

```

```

<400> SEQUENCE: 23

```

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ccttctaaga aagtgtggtg      20

```

-continued

<210> SEQ ID NO 24
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 24

ggatggagaa ttagcacgt 20

<210> SEQ ID NO 25
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 25

ctgactgact gactgactga ctattgtctc gttgettccg ttgtttc 47

<210> SEQ ID NO 26
<211> LENGTH: 49
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 26

gactgactga ctgactgact gactcagcag agcttggaag agagggatg 49

<210> SEQ ID NO 27
<211> LENGTH: 44
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 27

gactgactga ctgactagga tacagttcgg actcgaattg tcac 44

<210> SEQ ID NO 28
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 28

gactactaac tttaagccca cgaccca 27

<210> SEQ ID NO 29
<211> LENGTH: 38
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 29

actgactgac taaggatcta gtccttgcac aaataaca 38

<210> SEQ ID NO 30
<211> LENGTH: 56
<212> TYPE: DNA
<213> ORGANISM: Camellia sinensis

<400> SEQUENCE: 30

tgactgactg actgactgac tgactgactg acttagcatc taagaaagag gatttta 56

What is claimed is:

1. A method for evaluating tea plant caffeine content, comprising detecting a genotype of a molecular marker by using primers, wherein the primers consist of nucleotide sequences shown as SEQ ID NO: 8 and SEQ ID NO: 9, 5 wherein the molecular marker is a SNP site 3, wherein the SNP site 3 is located at position 501 of the nucleotide sequence of SEQ ID NO: 7, when a genotype of a sample is single mutant CT, the caffeine content in the dry matter in the tea plant is 10 more likely to be lower than the normal average of the sample of which the genotype is double mutant CC.

* * * * *